浙江大学本科生通识核心课程 - 控制论

稳定性
Stability

邵之江

浙江大学控制科学与工程学院

2023.10.17





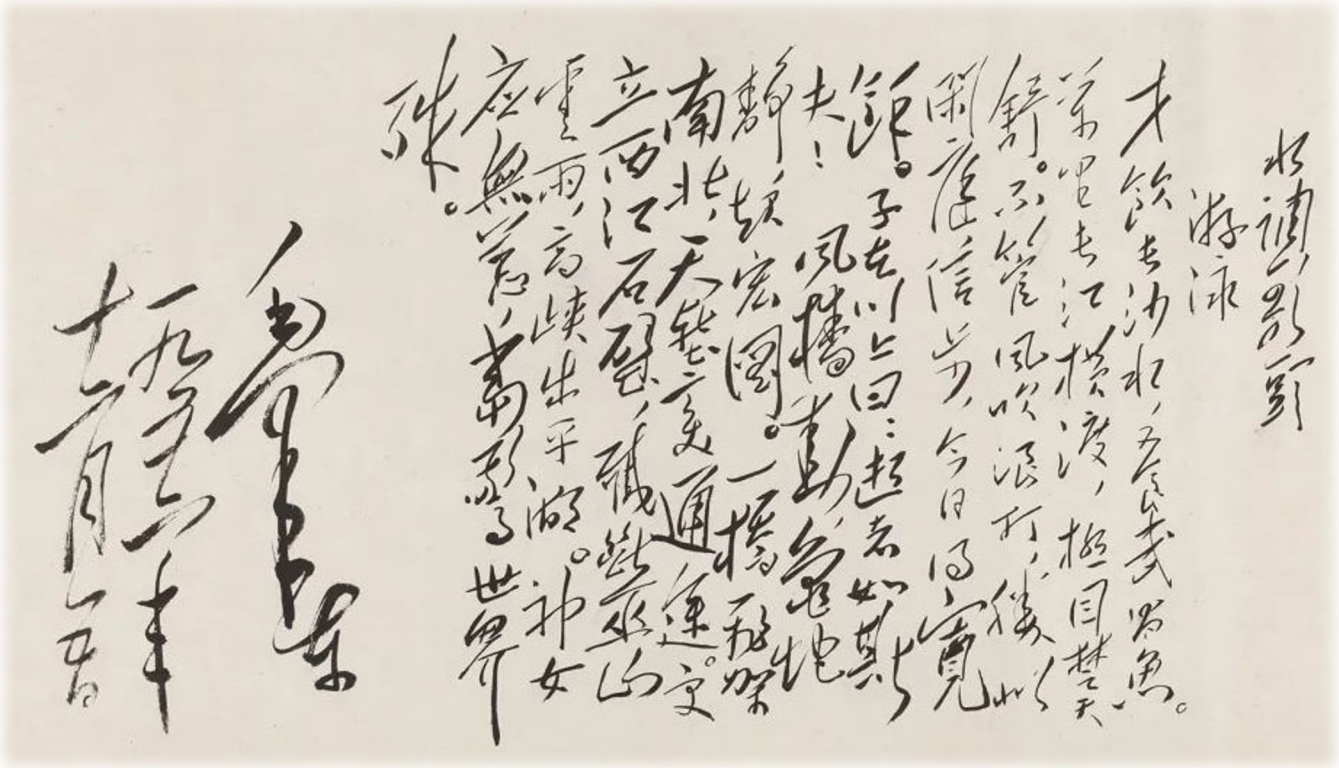
巴以冲突 - 不稳定的火药桶

加沙建筑



王毅：“以色列的行为已超越自卫范围，应认真倾听国际社会和联合国秘书长的呼吁，停止对加沙民众的集体惩罚”

水调歌头·游泳(毛泽东，1956. 6)



万里长江横渡
极目楚天舒
不管风吹浪打
胜似闲庭信步

稳！

“稳”字的基本信息：繁體 穩 簡 常 通

稳

拼音	注音	部首	字形分析	异体字
wěn	ㄨㄣˇ	部首 禾 部外 9 总笔画 14	左右结构 形声；从禾、急省声	穩㥯穩㝱㤷㥯㞗㥯穩㝱㝱躖

字源演变：



小篆 楷体

字形对比：



中国大陆 台湾 韩国 旧字形

异体字



穩 㥯 穩 㝱 㤷 㥯 㞗 㥯



穩 㝱 㝱 躖

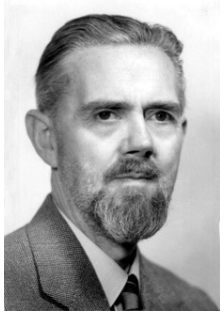
起源：粮食安全

守底线、固根本

判断力、定力
执行力、组织力

控制论导论(1956) - W.Ross Ashby 艾什比

- 普遍和抽象：类似于“几何学”：抽象形状
- 理想化的假设：类似于“物理学”：光滑…
- 统一概念：伺服系统 vs 小脑反射作用
- 复杂系统：克服单因素、局部分析、实验限制



- 控制论讲的不是物件而是动作方式……本质上并不依赖于物理定律和物质的性质
- 控制论研究一切形式的动态，只要它们是有规则的，或者说是规则的，或者说是可以再现的。控制论的真理不需要以其他科学为依据。控制论有它自己的基础

控制科学 CYBERNETICS

- 控制科学是一门“横断科学”(transversal discipline)

  动态、系统、调控、稳定、鲁棒、协同、适应.....



哲学

数学

控制科学

自动化技术



传感、检测
建模、分析
调控、优化
协同、执行

工程科学

矿产、建筑、交通、制造、航天、能源、材料、环境…

经济社会

货币、税收、投资、房地产、股票、服务业、政府……

生物医药

生命科学、医疗、生物、农业……

稳的词组

稳定、稳固：重心稳、底盘稳
稳重、稳健、稳当：立场稳、态度稳、操作稳
平稳、安稳：情绪平稳，经济运行平稳
稳妥、沉稳、稳扎稳打、稳坐钓鱼台、稳如泰山
作为动词，稳住：物理层面、精神层面

steady, stable, constant, firm, certain, sure…
balance, equilibrium, homeostasis…
relulation, stabilization, maintaining,

动态性的“稳”
系统性的“稳”



平衡之美 – 奇妙的稳定



NOTES AND DISCUSSIONS

Toys in physics teaching: Balancing man

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(Received 15 October 1985; accepted for publication 4 December 1985)

Balancing toys and other similar devices have long been used to discuss stability of equilibrium. A balancing toy horse and rider are illustrated in a text that is 125 years old,[1] and an almost identical device is available currently from a scientific supply house.[2] Balancing men of various sorts have been used as examples in physics texts for about as long.[3–6] In general, the texts discuss the location of the center of gravity of the system and how this leads to stable equilibrium, while on occasion a passing reference is made to oscillations about the equilibrium position. Closer examination of these oscillations can lead the student to additional insights about the motion of rigid bodies.

A toy balancing man is shown in Fig. 1.[7] This toy is most often used to illustrate stable equilibrium. While it appears to be precariously balanced (unstable equilibrium), in fact it is designed so that the center of gravity is below the pivot point, and tipping it in any direction raises this center of gravity. This results in a restoring torque, and hence stable equilibrium. The location of the center of gravity of the balancing man is readily determined. From symmetry, it must lie along the vertical axis of the toy. The actual point along the axis can be determined either by hanging the man from some point that is off axis and determining the intersection of a vertical line with the axis, or by balancing it across a straight edge and measuring the intersection of the straight edge with the axis. Measurements show the center of gravity to be 2.06×10^{-2} m below the balance point for this particular balancing man.

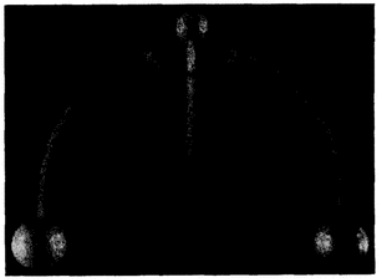


Fig. 1. A photograph of the balancing circus man (approximate dimensions are 0.17 m wide by 0.13 m high).

Additional ideas about rigid body motion can be illustrated by examining the motion of the toy when it is disturbed from its equilibrium. If the balancing man is tipped slightly and released, it will perform oscillations about the equilibrium. When it is tipped about an axis perpendicular to the plane of the toy (z axis) and then released, it oscillates with a period of $T_z = 1.13$ s. When it is tipped about an axis parallel to the plane (y axis), its period is found to be $T_y = 0.62$ s.

The differences in these motions can be understood by considering the balancing man to be a physical pendulum. The period of a physical pendulum can be calculated in the following way:

$$T = 2\pi(I/Mgd)^{1/2},$$

where M is the mass, g is the acceleration of gravity, d is the distance between the balance point and the center of gravity, and I is the moment of inertia about the axis of interest. Since the balancing man has different moments of inertia about the y and z axes, it will have different periods of oscillation about these axes.

The total mass of the balancing man was measured to be 0.2717 kg. To calculate the moment of inertia about an axis, both the dimensions and the masses of the individual parts has to be determined. The masses were calculated from the measured volumes of the individual parts and the average density of the toy. This assumed that a uniform plastic was used throughout. Once the masses of the parts were determined, the calculation of the moments of inertia became an interesting exercise using known moments of inertia of standard objects, and the parallel axis theorem. As an example, the moment of inertia about the z axis was calculated from the following:

Spheres: $2[(2/5)mr^2 + mh^2]$, h = center of sphere to pivot;
Head: $(2/5)mr^2 + mh^2$, h = center of head to pivot;
Semicircle: $mr^2 + mh^2$, h = center of circle to pivot;
Body: $(1/12)mL^2 + m(L/2)^2 = (1/3)mL^2$;
Arms: $2[(1/12)mL^2 + mh^2]$, h = center of arm to pivot.

The moment of inertia about the z axis was then calculated as the sum of these individual moments. The calculated moments of inertia about three perpendicular axes were found to be

$I_z = 1.804\times10^{-4}$ kg m^2,
$I_y = 0.536\times10^{-4}$ kg m^2,
$I_x = 1.267\times10^{-4}$ kg m^2.

The moments of inertia were then used to calculate the periods of oscillation of the balancing man. The results

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平衡之美 – 不可思议的稳定



平衡舞大师志田美代子

稳定性

稳定压倒一切：重要性、表现方式

- 空调温度的稳定性
- 飞行过程的稳定性
- 骑自行车的稳定性
- 单腿站立的稳定性
- 电脑系统的稳定性
- 火箭发射/回收的稳定性
- 情绪的稳定性、工作的稳定性
- 生态系统的稳定性
- 社会系统的稳定性：政治、经济



共同特征：
抗扰、内稳
定性/定量

稳定的多重含义：结构设计、运行调控

- 设计、结构，由初始状态决定
- 运行、操作，在过程中不断调整



上海中心

大楼增稳的例子：上海中心

1、轮廓外形设计，流线型螺旋扭转，降低风荷载

2、阻尼减震控制，摆式电涡流调谐质量阻尼器，重量1000吨，极限摆幅2米

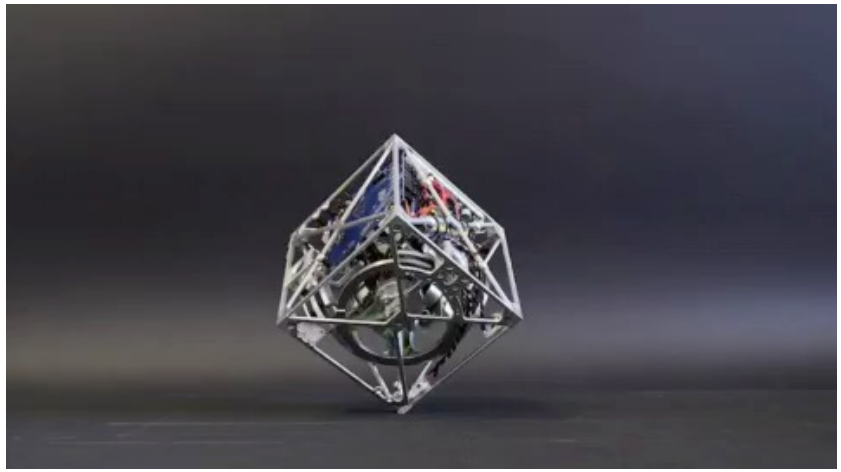


Tuned Mass Damper

稳定的多重含义：定值

定值：克服外部干扰、保持不变
设定值为固定连续取值，如房间温度
单腿站立、保持直立姿态

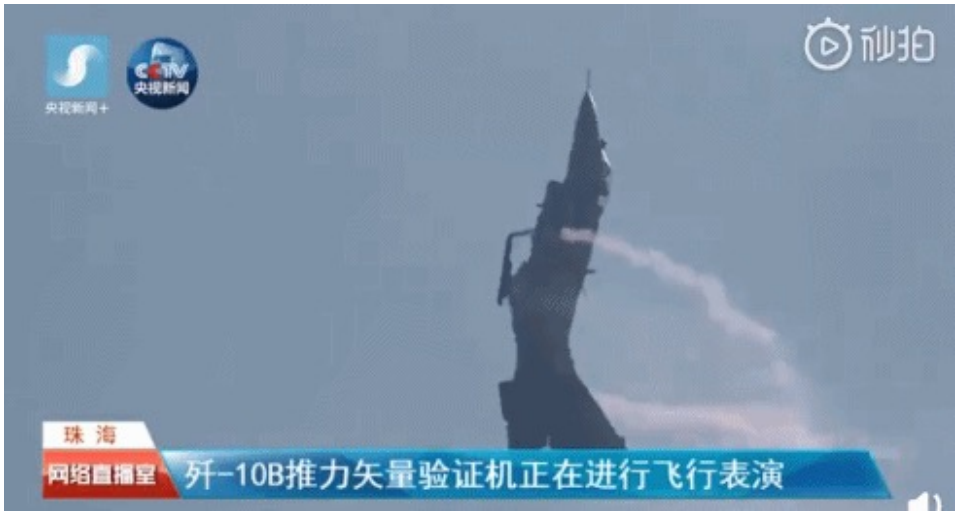
外部不确定性、内部不确定性
反馈调节机制



稳定的多重含义：随动

随动：轨迹跟踪、波动可控
运动控制：飞行控制、汽车驾驶

环境变动，不确定性、设定值的变化范围和变化速度
一阶导数、二阶导数、三阶导数

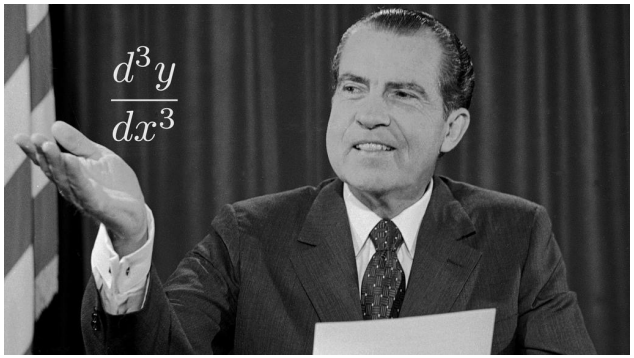


- First **position** x,
- then **velocity** $v = x' = \frac{dx}{dt}$,
- then **acceleration** $a = x'' = \frac{d^2x}{dt^2}$,
- then **jerk** $x''' = \frac{d^3x}{dt^3}$,
- then **jounce/snap** $x'''' = \frac{d^4x}{dt^4}$,
- then **crackle** $x''''' = \frac{d^5x}{dt^5}$,
- then **pop** $x'''''' = \frac{d^6x}{dt^6}$,

稳定的多重含义：广义

广义：结构功能完整、性能行为符合预期、系统健康

完整性：多量化指标、整体性“功能”指标
平衡性：生态系统的稳定性，力量对比
长期性：效用持久、长时间有效、可用性好
扩散性：价格和社会影响
鲁棒性、适应性...



the rate of increase of inflation was decreasing, 1972, Nixon



the perfect image stabilizer

内稳定Homeostasis

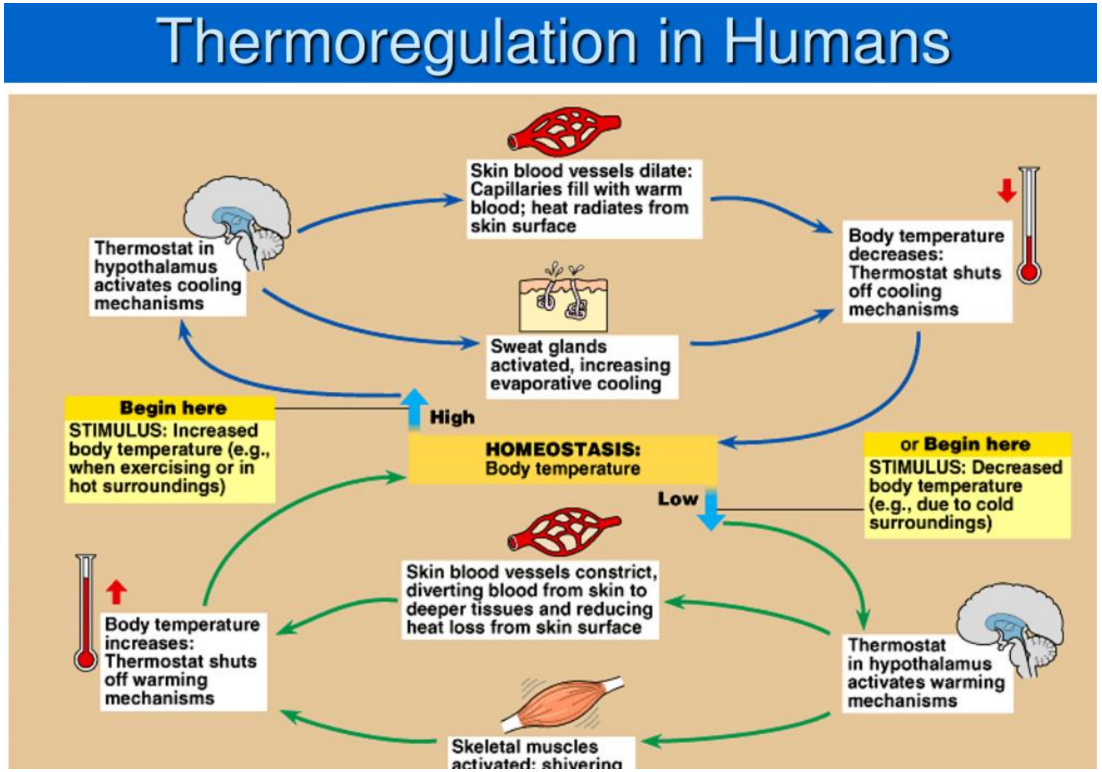
Homeostasis, from the Greek words for "same" and "steady," refers to any process that living things use to actively maintain fairly stable conditions necessary for survival.

- internal environment vs external environment
- balance, equilibrium, homeostasis
- relulation, stabilization, maintaining, self-government.



William Bradford Cannon, 1871-1945

The Wisdom of the Body. 1932



生命体中的控制

- Pancreas胰腺

  Regulates blood glucose葡萄糖 level
- Adrenaline肾上腺素

  Automatically generated to increase the heart rate and oxygen in case of fear, excitement, nervousness, ... (gives you more energy when in emergency)
- Eyes and Hands
  - Use eyes to follow moving object
  - Use hands to pick up an object and place it at a predetermined location
- Body Temperature
  - Skin and Blood systems
  - Regulated temperature of 36° C to 37° C
  - Fever against bacteria and viruses

《控制论》，page75

一个病人到神经病院来临诊，他并没有瘫痪，当他接到命令时还能移动下肢。然而，他苦于严重的病废，他在步行时呈现特殊的不准确的步态，两眼朝下，看着地面和足部。他每走一步腿都抬得很高，而且迈足过度，上身则落在后而，如果遮住他的两眼，他就要站立不住而踉跄倒地，这是怎么回事呢？

另一个病人也来临诊，当他安静地坐在椅中时，好像没有什么毛病。但是，如果给他一支香烟，在他企图接取这支烟时，他的手会摇摆不定而抓不到它。接着他的手又在另一侧作同样无益的摇摆，随后，第三次又摇摆回来，他的手一直就这样进行着无益而激烈的振荡。如果给他一杯水，当他把这杯水端到口边以前就会由于这些摇摆而泼空了。这又是怎么回事呢？

脊髓痨，运动神经失效，信息感知和传递受损

目的性震颤，小脑受损

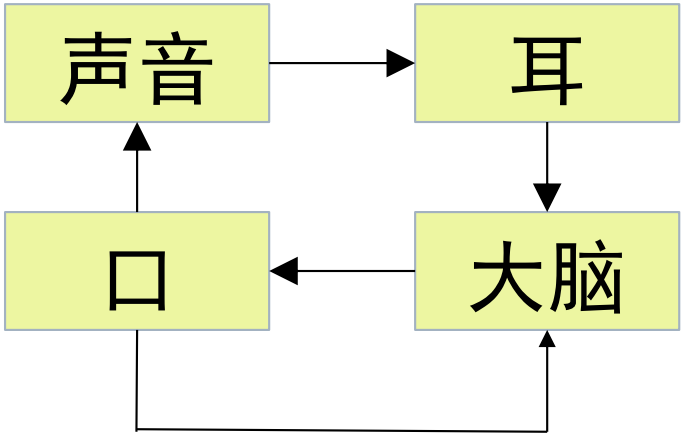
不足的反馈、过度的反馈，都会妨碍“有组织的活动”-维纳

七种震颤 Tremor

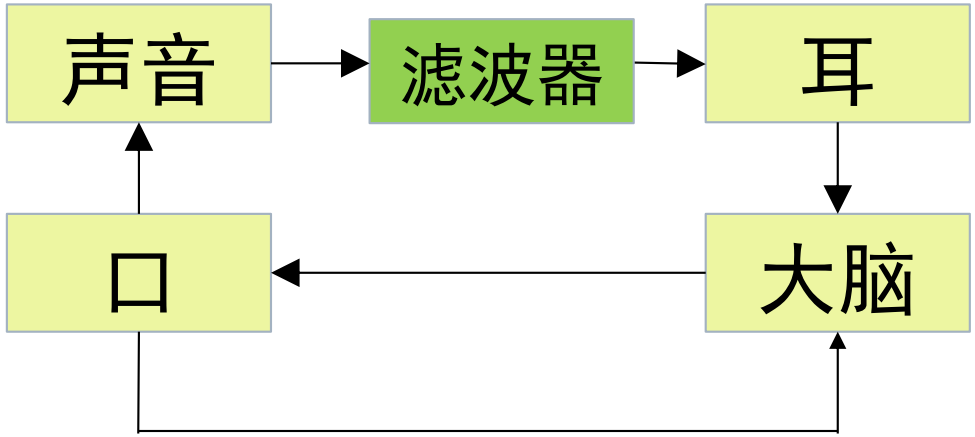
1、原发性震颤，与任何已知疾病无关，相对温和，机理不明。家族性、遗传性？
2、肌张力障碍性震颤，一种运动障碍，由于大脑的不正确信息传递导致肌肉过度活动，出现异常的姿势或持续的、非意愿的运动
3、小脑性震颤，四肢(例如手臂、腿)的缓慢且振幅较大的震颤，它是由小脑及其他脑区损伤引起的。损伤可能会由疾病所致，如中风、肿瘤、多发性硬化症或遗传性神经退行性疾病（如共济失调、脆性X综合征等），长期饮酒也会造成小脑的慢性损伤
4、心因性震颤，潜在的精神障碍，如抑郁症或创伤后应激障碍（PTSD）
5、生理性震颤，发生在所有健康人身上。它几乎是肉眼不可见的，也不被认为是一种疾病，而是一种正常的生理现象或身体的物理特性（节奏规律的活动如心跳和平滑肌反应）
6、 直立性震颤，站立时腿部肌肉迅速收缩或痉挛。使患者有不稳定或不平衡的感觉，并立即尝试坐下或行走。原因不明
7、帕金森震颤，一只或两只手在休息时发抖，并且也会影响到下巴、嘴唇、脸部和腿部。身体一侧或两侧。会因紧张或强烈的情绪而变得更严重

口吃和纠正

- 非自愿的重复、停顿、拖长
- 语言表达过程的失调、失控
- 自我紧张、恶性循环
- 滤波器：减弱反馈回路作用



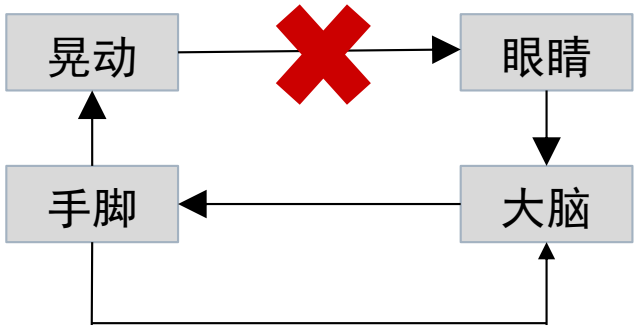
讲话过程的两个回路



用滤波器减弱反馈

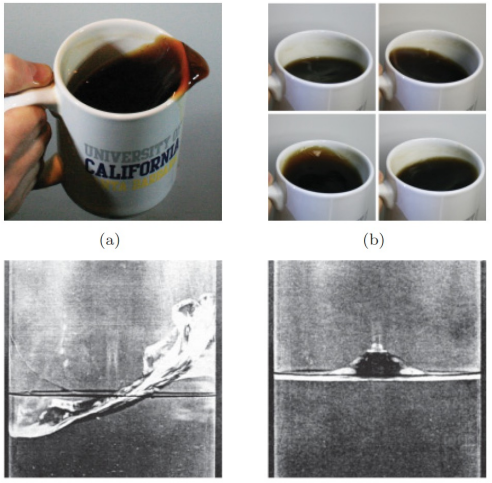
端咖啡杯走路

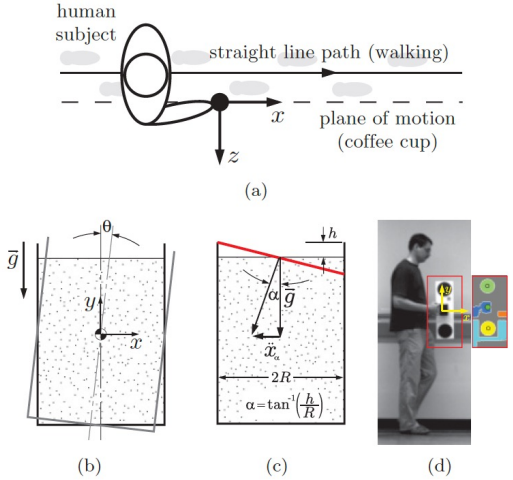
目标：平稳、快速、不泼溅
分析：结构、节奏、共振、心理因素
反馈调控：紧张过度、反馈切除



H. C. Mayer and R. Krechetnikov, Walking with coffee: Why does it spill? Phys. Rev. E **85**, 046117, 26 April 2012

- cup sizes and initial liquid levels
- coffee properties and low-viscosity-liquid dynamics
- walking speeds and biomechanics





higher-frequency harmonics. Comparison of the focused and unfocused regimes in Fig. 5 suggests that higher levels of noise in the latter regime substantially intensify the spilling.

B. Mechanical model

To support the above conclusions and explain the experimental findings presented here, we developed a simple mechanical model based on the fully nonlinear spherical pendulum equations with forced oscillations [23]. As a spherical pendulum (along with spring-mass models [24]) have characteristics similar to sloshing, several investigators have used this analogy to predict and understand the observed sloshing behavior in other systems [8,25–27].

Since we are mainly interested in the near-planar pendulum motions, it is convenient to choose a coordinate system which has singularities at $x = z = 0$, $y = \pm 1$ (the points on the sphere that are furthest from the plane) and is defined in terms of two angles γ and β:

$$x = \cos\beta \sin\gamma, \quad y = \sin\beta, \quad z = -\cos\beta\cos\gamma, \quad (2)$$

where $-\pi < \gamma \leqslant \pi$ and $-\pi/2 \leqslant \beta \leqslant \pi/2$ as shown in Fig. 6. Considering a spherical pendulum of length l and mass m, which is forced horizontally in the x direction as $x_0(t)$, the nondimensional kinetic and potential energies are given by

$$\frac{T}{m\,g\,l} = \frac{1}{2}[(\dot{x} + \dot{x}_0)^2 + \dot{y}^2 + \dot{z}^2], \quad \frac{V}{m\,g\,l} = z, \quad (3)$$

where we took into account that the horizontal displacements of the pendulum relative to the pivot are lx and ly, respectively. Given the expressions of the kinetic and potential energies, we can form the Lagrangian $L = T - V$ and, expressing x and z in terms of y and γ, derive the Euler-Lagrange equations [29] for γ and y:

$$\ddot{\gamma} - \frac{2\,y\,\dot{y}\,\dot{\gamma}}{1-y^2} + \frac{\sin\gamma}{\sqrt{1-y^2}} = -\frac{\ddot{x}_0\cos\gamma}{\sqrt{1-y^2}},$$

$$\ddot{y} + \frac{y\,\dot{y}^2}{1-y^2} + y(1-y^2)\dot{\gamma}^2 = y\sqrt{1-y^2}\,(\ddot{x}_0\sin\gamma - \cos\gamma).$$

The length l of an equivalent spherical pendulum is found by equating the natural frequency of the pendulum to that of the first antisymmetric mode ω_{11} [cf. Eq. (1)], for liquid sloshing

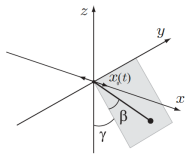


FIG. 6. Coordinate system for the spherical pendulum model: the pendulum is shown in the shaded plane and driven horizontally in the x direction with time-dependent forcing $x_0(t)$.

控制论看疫情防控和社会治理

目标、反馈、决策、执行
稳定性：能否经受得住大风大浪

- 集中统一领导、全国一盘棋，全社会总动员
- 未雨绸缪、前瞻布局，系统谋划
- 令行禁止、联动协同、群防群治
- 集中资源、稳定民心：医疗资源、航空、铁路、公路、主要工业品、粮油等各类保障物资
- 产业基础和人才优势：生产能力、配套能力、应变能力、组织能力

控制论看疫情防控和社会治理

疫情防控 VS 经济社会秩序恢复

- 超级复杂系统：交通、社区、学校、企业、市场、娱乐...
- 决策难度：系统嵌套耦合、对象动态多变、潜伏滞后...
- 全方位协同：资源、信息、决策、手段
- 稳定：医疗物资、生活物资、全社会资源...人心第一！
- 信息角度：精确、精准、精细

1. 隔离划分：分区分级分层、密度控制、分布式管控
2. 情报监测：关键变量、关键指标、大数据融合
3. 系统决策：综合决策施策、过调过激问题
4. 治病救人：机理、疫苗、执行力、公卫体系

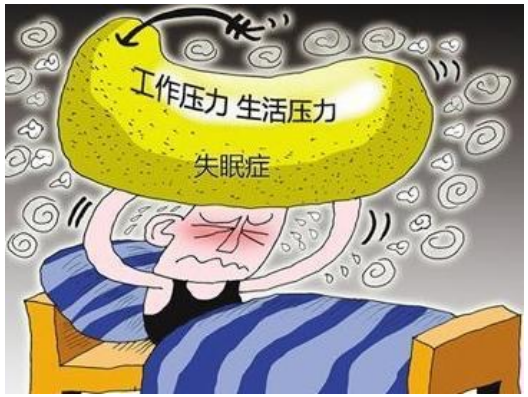
生活中的稳定和不稳定

决策和执行往往是离散事件，而非连续调控过程

例子：和面过程的调控和干湿振荡
水温/配比/添加剂/步骤/手法…

例子：睡眠过程的干预和失眠振荡

例子：人际关系处理



调节过程的感知、反馈、实施，形成闭环

生活中的稳定和不稳定

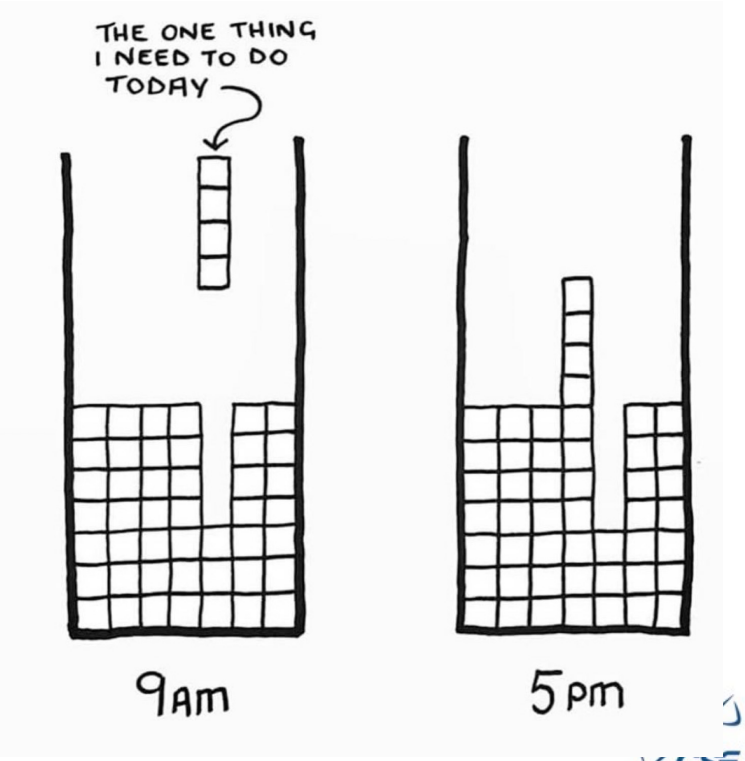
有目标、有调控，但：

南辕北辙、刻舟求剑、适得其反、震荡反复

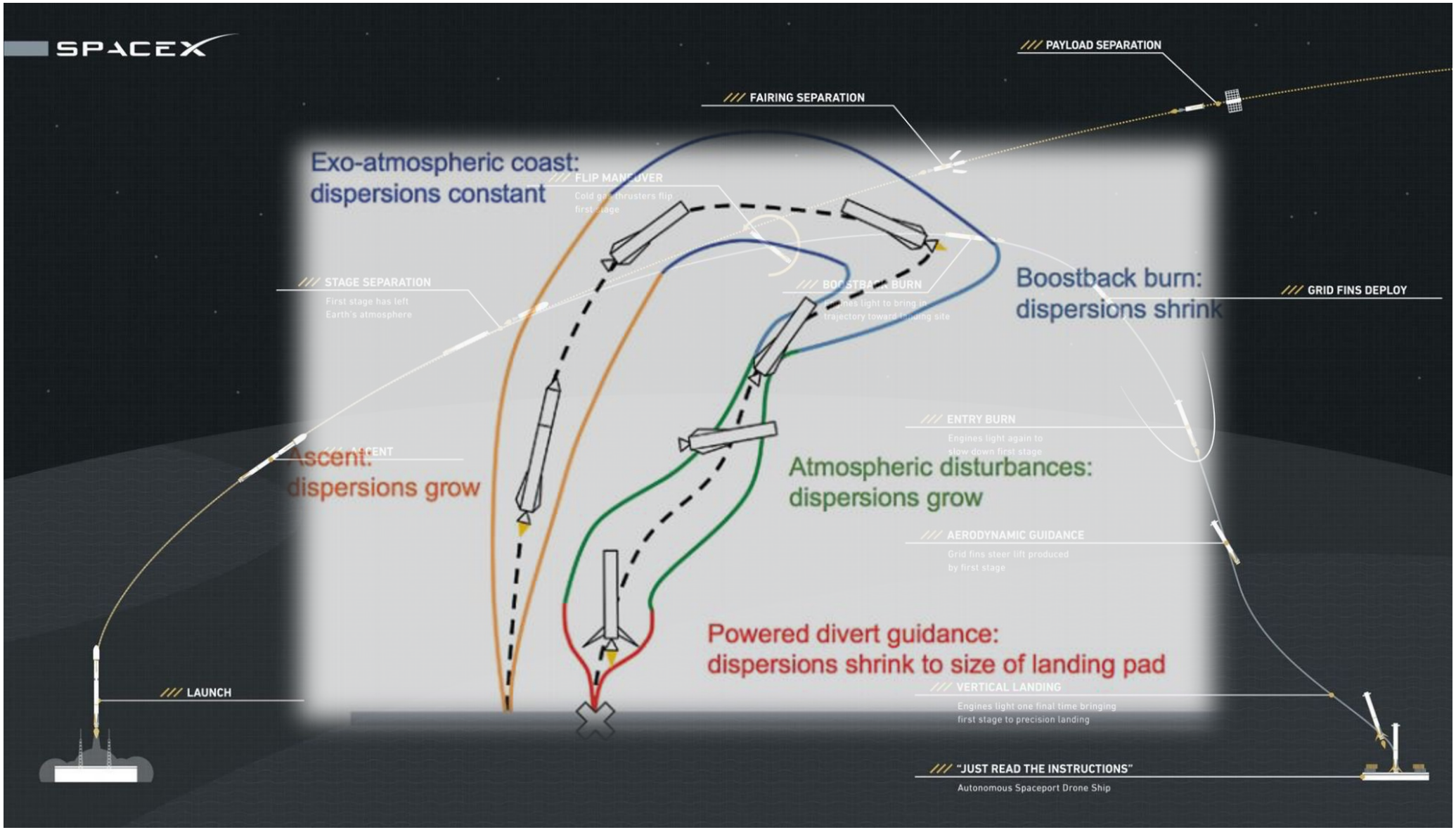
反复折腾、平衡破坏、不可逆失效

有目标、有手段、有反馈的前提下，如何解决稳定性的难题？

打破！跳出！重构！



VTVL


SPACEX
PAYLOAD SEPARATION
FAIRING SEPARATION
Exo-atmospheric coast:
dispersions constant
FLIP MANEUVER
STAGE SEPARATION
First stage has left
Earth's atmosphere
Boostback burn:
dispersions shrink
GRID FINS DEPLOY
ENTRY BURN
Engines light again to
slow down first stage
Ascent:
dispersions grow
Atmospheric disturbances:
dispersions grow
AERODYNAMIC GUIDANCE
Grid fins steer lift produced
by first stage
Powered divert guidance:
dispersions shrink to size of landing pad
VERTICAL LANDING
Engines light one final time bringing
first stage to precision landing
LAUNCH
"JUST READ THE INSTRUCTIONS"
Autonomous Spaceport Drone Ship

Falcon9 Landing Failures







稳定性：从定性到定量

- 系统：边界、环境、目标、约束
- 划定表征系统行为的变量(特征量)范围，没有跑到外面
- 范围的定义：一个圆圈、一个带状区域(随时间变化)、一个功能的组合/序列/网络
- 现实：在外界干扰下，系统的行为发生波动，会产生不期望的效果，或者可能产生不了预期的效果
- 调控：输入、输出、特性、规律、干扰、噪声

稳定性：判断、任务

判断

1、波动的范围是否满足要求？

BIBO Bounded-Input Bounded Output

2、如果外界干扰消失了，系统能否恢复？

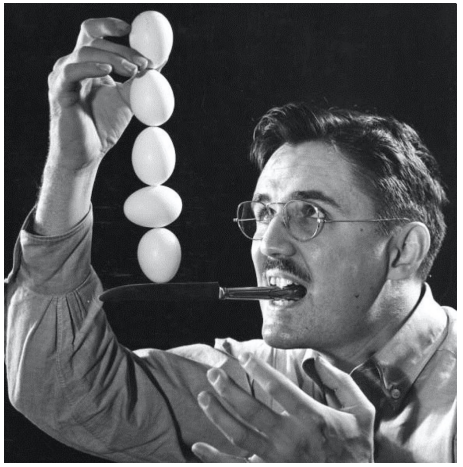
分类：

1、简单任务：单输入单输出、无干扰无噪声

2、复杂任务：复杂对象特性、复杂任务要求、复杂外部环境、复杂控制系统(新场景、新问题：设计错误、性能退化、安全防范…)

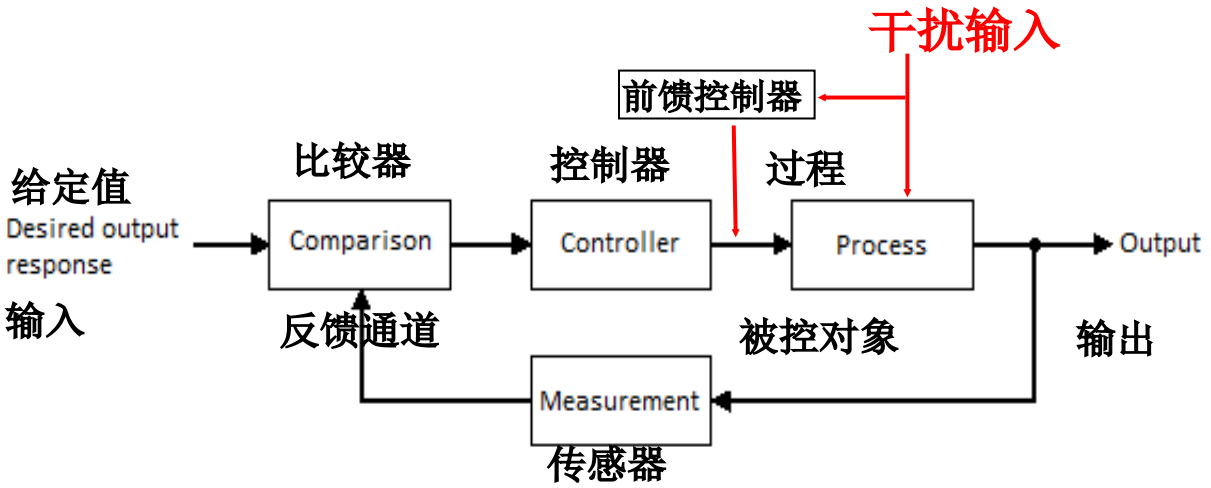
稳定性：不控、失控

不稳定：定义、现象、原因、消除办法
竖铅笔、竖硬币、竖石头、竖鸡蛋...

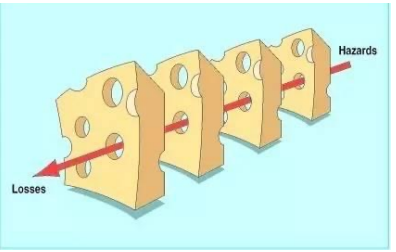


egg balancing

不稳定的两种情形：
不控：自稳定范围/裕度、外扰/环境
失控：负反馈/正反馈、过度调节
失控的根源：部件问题、节奏问题、导向问题

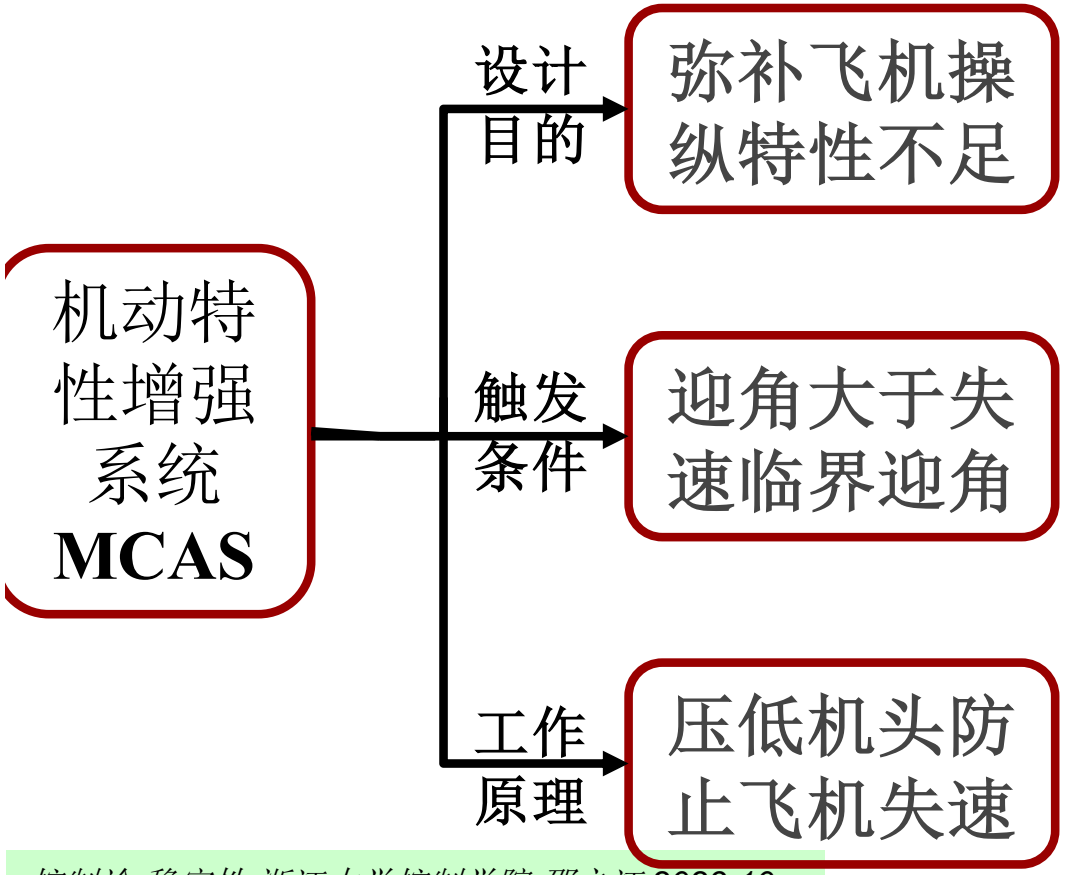


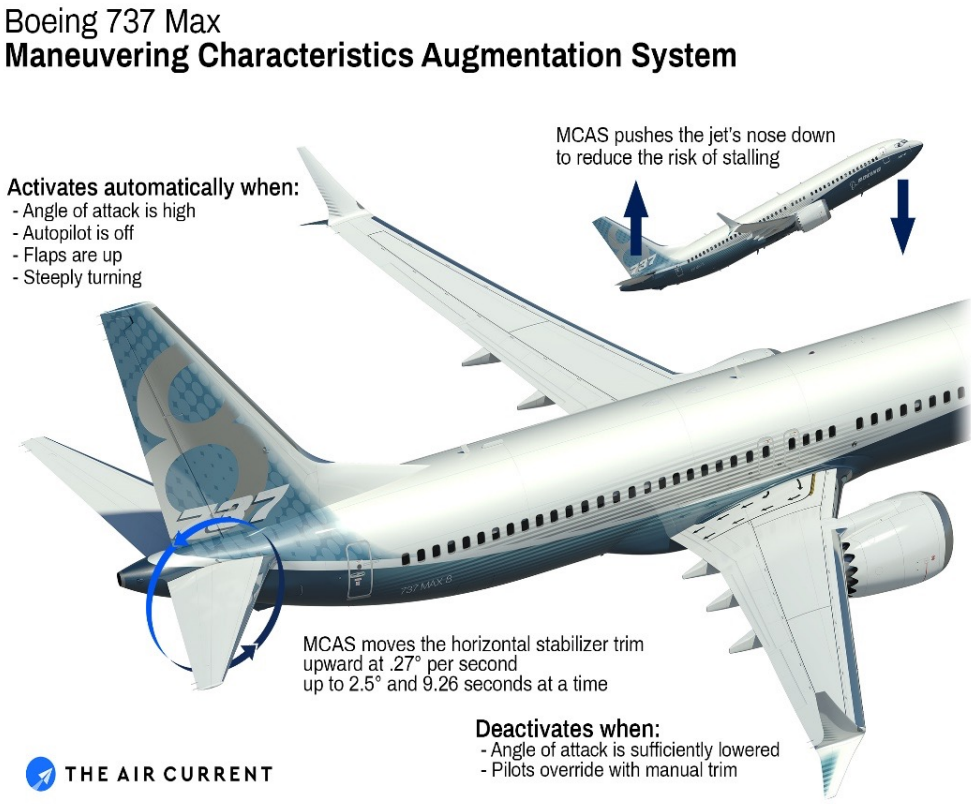
失控分析：波音787MAX事件



2019.3.10，埃塞俄比亚航空公司，157人遇难
2018.10.29，印度尼西亚狮航，189人遇难

瑞士奶酪模型：发动机、传感器、智能驾驶仪



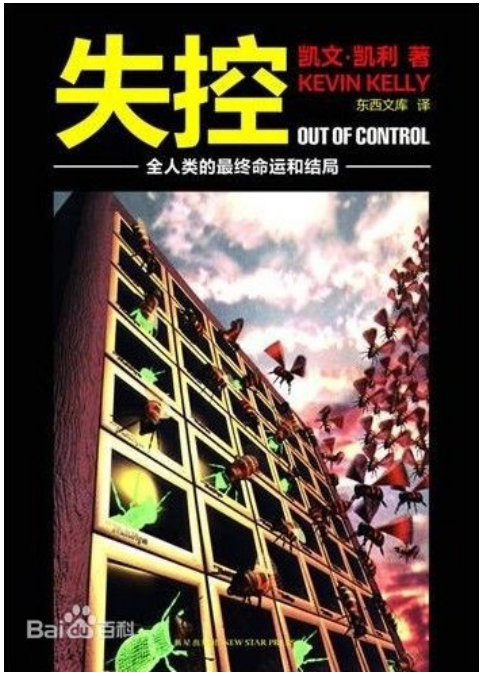


《失控》

《失控：机器、社会与经济的新生物学》

Out of Control: The New Biology of Machines, Social Systems, and the Economic World，Kevin Kelly，1994





造物九律：分布式状态、自下而上的控制、培养递增收益、模块化生长、边缘最大化、礼待错误、不求目标最优、但求目标众多、谋求持久的不均衡和变自生变

上德不德，是以有德。下德不失德，是以无德 – 道德经

与稳定性相关的概念

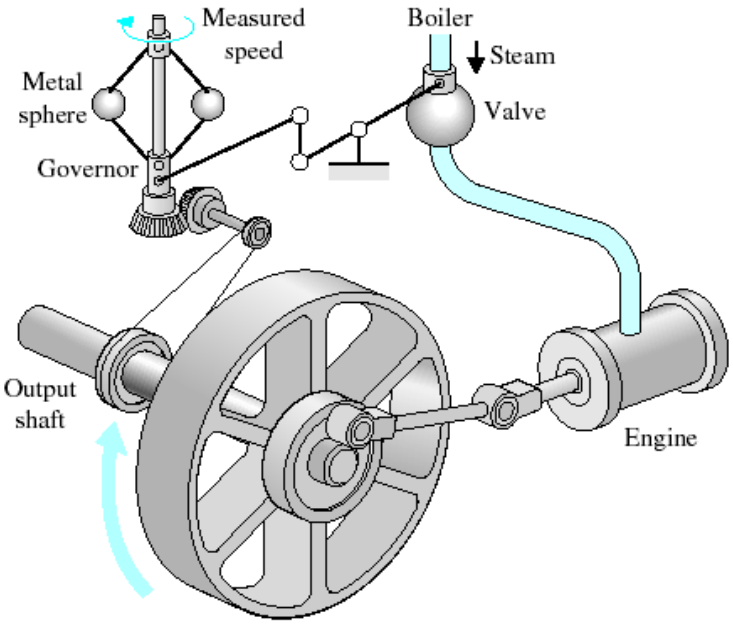
重复性　　适应性

精确性　　鲁棒性

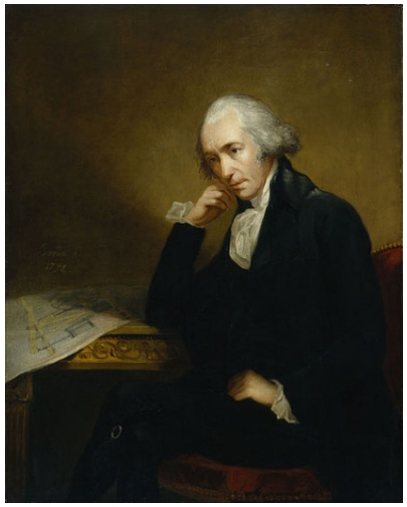
完整性　　可靠性

一致性　　智能性

稳定性分析的需求：蒸汽机调速



Watt's Flyball Governor
(18th century)

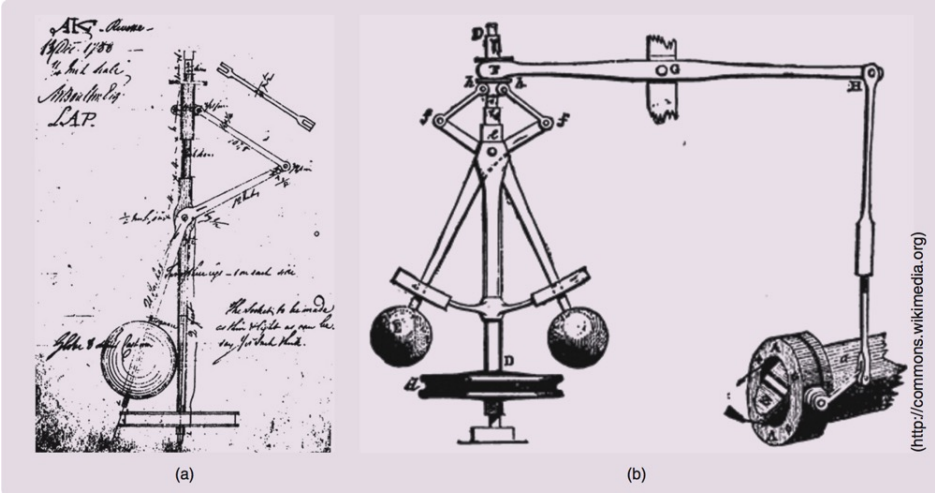


James Watt (1736–1819)

大工业时代复杂装置的调速问题
- an offset (steady-state error)
- low power of actuator
- hunting,or oscillation problem
- nonlinearity

连续反馈→修正偏差→达到目标→参数可调

稳定性的理论：麦克斯威尔

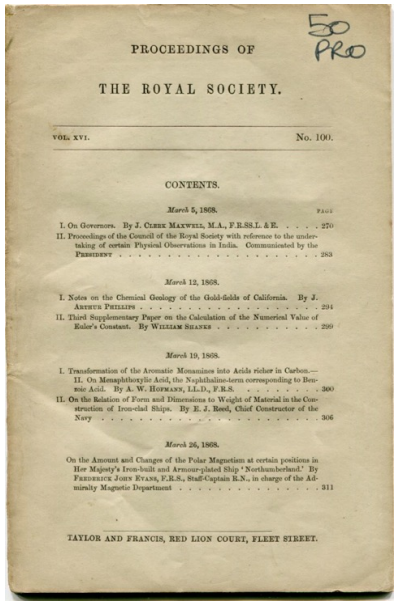




James Clerk Maxwell
(1831–1879)
Electromagnetism

“Father of control theory”

J. C. Maxwell,
“On governors,”
Proceedings
Royal Society of
London, vol. 16,
pp. 270–283,
Mar. 1868.



PROCEEDINGS OF

THE ROYAL SOCIETY.

VOL. XVI. No. 100.

CONTENTS.

March 5, 1868.

I. On Governors. By J. Clerk Maxwell, M.A., F.R.S.S.L. & E. 270

II. Proceedings of the Council of the Royal Society with reference to the undertaking of certain Physical Observations in India. Communicated by the President 283

March 12, 1868.

I. Notes on the Chemical Geology of the Gold-fields of California. By J. Arthur Phillips 294

II. Third Supplementary Paper on the Calculation of the Numerical Value of Euler's Constant. By William Shanks 299

March 19, 1868.

I. Transformation of the Aromatic Monamines into Acids richer in Carbon.—II. On Monaphthoxylic Acid, the Naphthaline-term corresponding to Benzoic Acid. By A. W. Hofmann, LL.D., F.R.S. 300

II. On the Relation of Form and Dimensions to Weight of Material in the Construction of Iron-clad Ships. By E. J. Reed, Chief Constructor of the Navy 306

March 26, 1868.

On the Amount and Changes of the Polar Magnetism at certain positions in Her Majesty's Iron-built and Armour-plated Ship 'Northumberland.' By Frederick John Evans, F.R.S., Staff-Captain R.N., in charge of the Admiralty Magnetic Department 311

TAYLOR AND FRANCIS, RED LION COURT, FLEET STREET.

稳定性的理论：麦克斯威尔

whole apparatus. The solution is of the form

$$x = A_1 e^{n_1 t} + A_2 e^{n_2 t} + A_3 e^{n_3 t} + Vt, \quad . \quad . \quad . \quad . \quad . \quad . \quad . \quad (11)$$

where n_1, n_2, n_3 are the roots of the cubic equation

$$MBn^3 + (MY + FB)n^2 + FYn + FG = 0. \quad . \quad . \quad . \quad . \quad (12)$$

If n be a pair of roots of this equation of the form $a \pm \sqrt{-1}b$, then the part of x corresponding to these roots will be of the form

$$e^{at} \cos (bt + \beta).$$

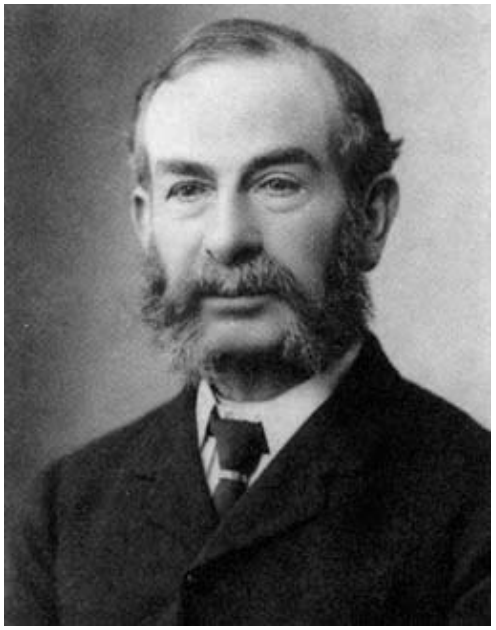
If a is a negative quantity, this will indicate an oscillation, the amplitude of which continually decreases. If a is zero, the amplitude will remain constant, and if a is positive, the amplitude will continually increase.

1. stability of a dynamic system by using a linear differential equation of motion
2. obtained a specific stability condition $a_1a_2>a_3$ for a linear differential equation of the third order from the solutions of its characteristic equation $s^3+a_1s^2+a_2s+a_3 =0, a_i>0$
3. introduced the linearization of a nonlinear differential equation
4. classified the motion of a machine into four kinds in view of the stability of a linear system
5. proposed an unsolved problem to determine stability conditions for a linear differential equation of the nth order.

非线性系统线性化
特征方程的多项式的系数
三阶系统稳定性充要条件

稳定性的理论：劳斯

不对特征方程进行求根，直接根据系数构造劳斯阵列，判定是否存在正实部(不稳定分量)



Edward Routh
1831-1907
born in Quebec
English mathematician

Fellow of the Royal Society, 1872

在计算机尚未诞生的年代，绝顶聪明的方法

Routh, E. J. (1877). *A Treatise on the Stability of a Given State of Motion*. MacMillan And Co.

稳定性的理论：李雅普诺夫

The general problem of the stability of motion,
Moscow University on September 12, 1892

一般非线性动态系统的稳定性判据：将解的稳定性与某个能量函数的存在性相关联，奠定了稳定性理论分析的基石

稳定性分析理论：
逻辑闭环，顶天立地
抽象通用，跨界复制
数学支撑，完美解决



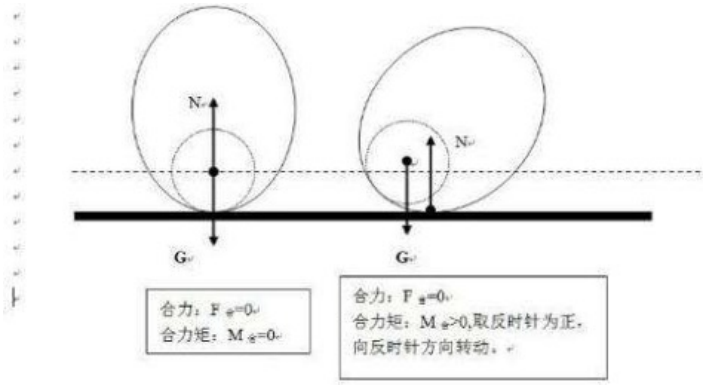
Aleksandr Mikhailovich Lyapunov
1857.6.6 - 1918.11.3
Yaroslavl, Russian Empire

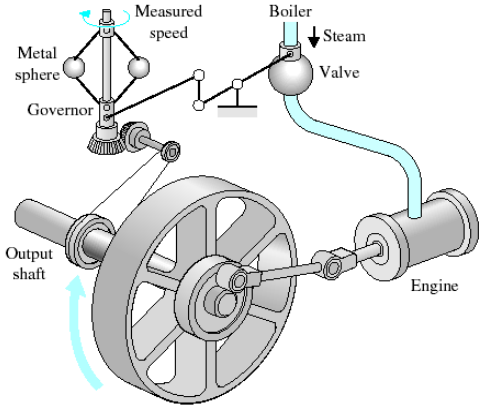
Applied mathematics
Saint Petersburg State University
Russian Academy of Sciences

增强稳定性的做法

物理结构的反馈，纯硬件实现，振荡，但无能耗、无通讯、无计算，非常可靠









机械+电子+陀螺仪
+控制算法+AI



增强稳定性的做法

1、增强感知和手段：因果性、必然性

2、壁垒、缓冲：阻断、阻尼

3、积极灵巧反馈：及时动作，克服不确定性

4、增强对干扰和变化的预见性

平衡：平稳性和快速性

稳定：因果链和反馈链的相对统一

调节手段、调节策略、调节效果、调节的调节...

关于稳定性的总结

1、系统、整体：抗外部干扰，克服内部不确定性

2、功能完整，行为可预测、效果符合预期

3、不足的反馈、过度的反馈，都会产生不稳失控

4、从定性到定量，精巧设计需要基于模型

5、广义：底线思维、智慧调控、信心定力

讨论题(开放选题)

1、讨论政治、经济、生态领域的稳定性问题

2、成语、谚语中的控制及稳定性

3、试分析不倒翁的稳定性

4、站立(蒙眼/单腿/平衡木/…)的稳定性分析

5、不控/失控、不稳定/稳定的辩证关系和案例分析

6、其他任何涉及稳定性的有趣话题

高能硬核

供选学进阶

案例1 - 单摆

物理模型

↓

（非线性）微分方程模型

↓

求解平衡点

↓

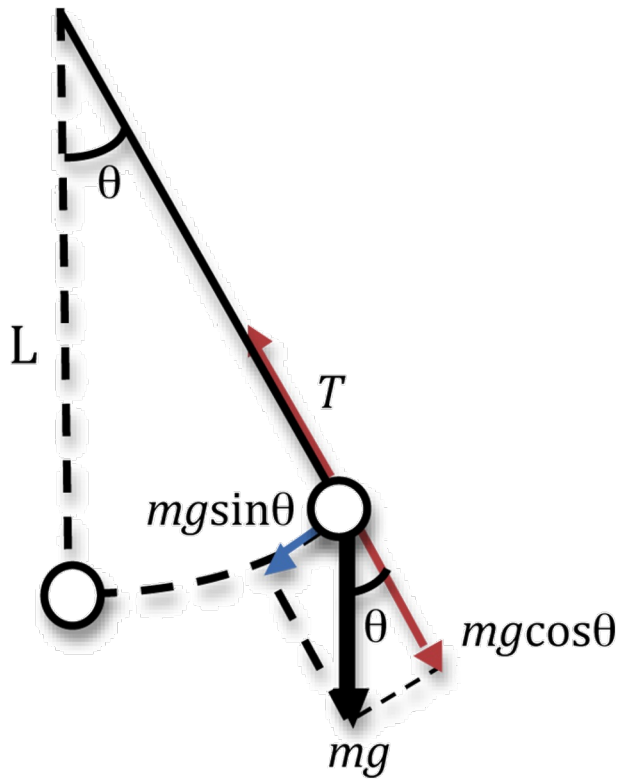
李雅普诺夫稳定性判定

↓

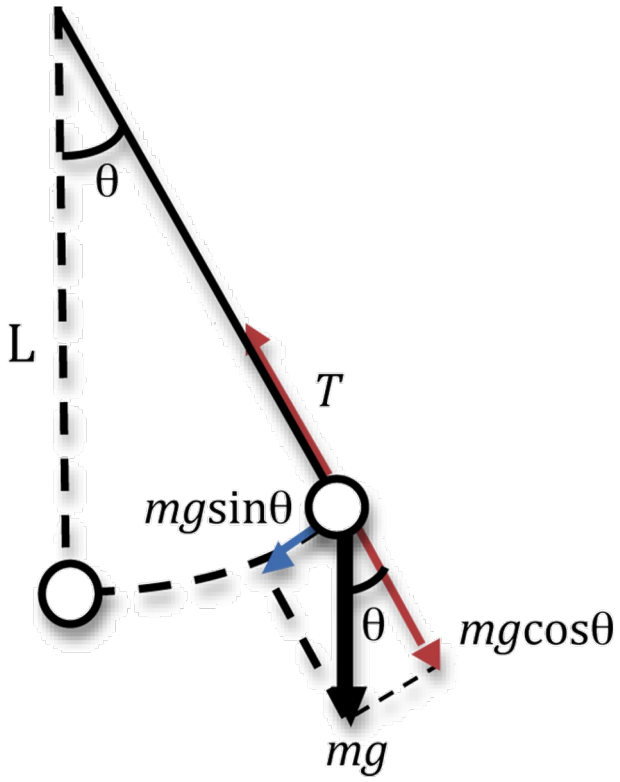
线性化

↓

稳定性判定



动力学建模（非线性）



θ：摆杆与竖直方向夹角

ω：摆球线速度

L：等效摆杆长度

b：空气摩擦系数(b<0)

g：重力加速度

系统微分方程：

$$\dot{\theta} = \omega$$

$$\dot{\omega} = -\frac{g}{L}\sin\theta + b\omega$$

稳定性

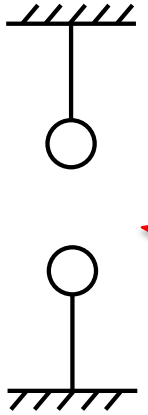
- **平衡点**

  对于自治系统$\dot{x} = f(x)$，使$\dot{x} = 0$的点称为平衡点

- 求解**平衡点**：

$$0 = \dot{\theta} = \omega$$
$$0 = \dot{\omega} = -\frac{g}{L}\sin\theta + b\omega$$

$$\Rightarrow \begin{cases} (\theta, \omega) = (0, 0) \\ (\theta, \omega) = (\pi, 0) \end{cases}$$



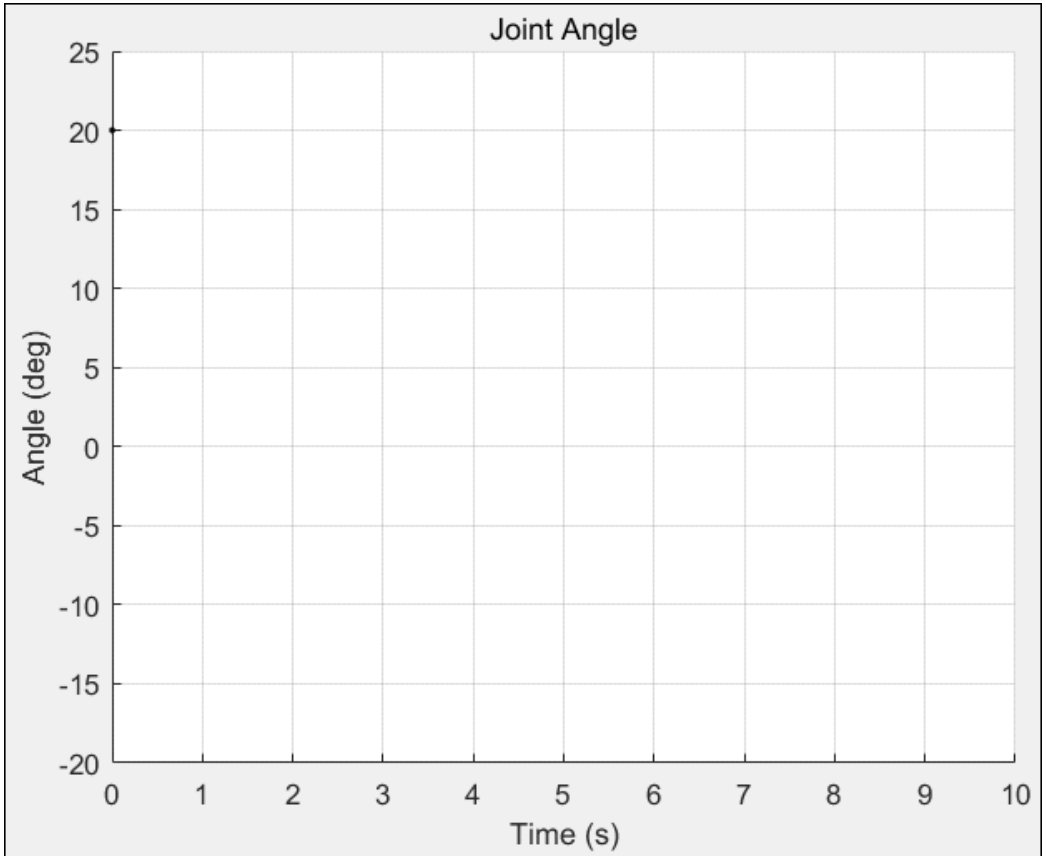
- **稳定性**定义：系统受外界干扰偏离平衡点。稳定性就是指扰动消失后，系统可以恢复到平衡点的性能。

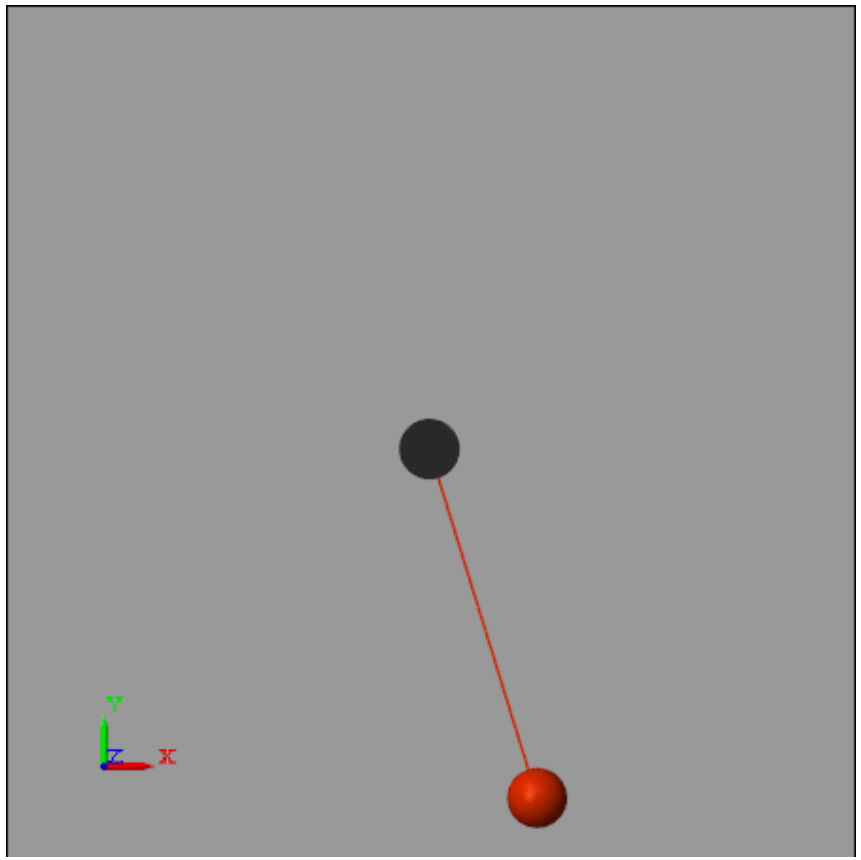
  注：控制理论中仅讨论**输入为零**，初始状态不为零时的稳定性。

思考：这两个平衡点稳定吗？

$(\theta, \omega) = (0, 0)$

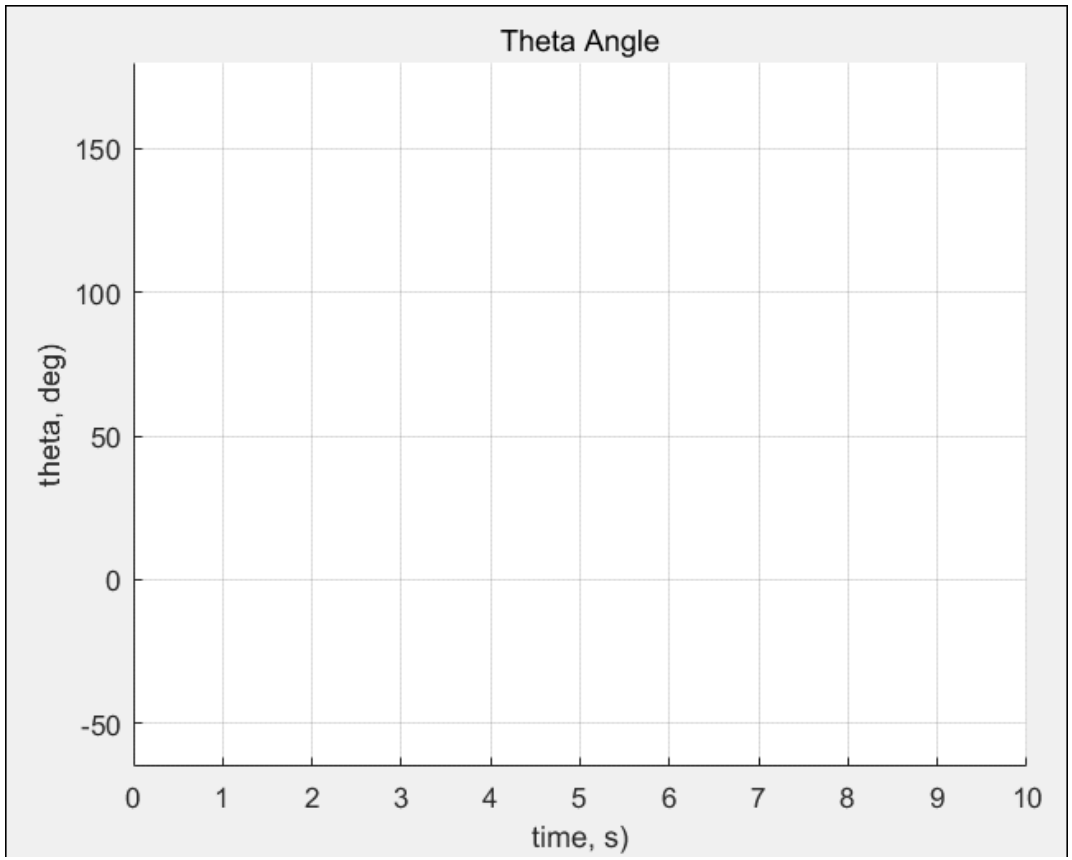
单摆仿真例子：　　sm_single_pendulum_sl

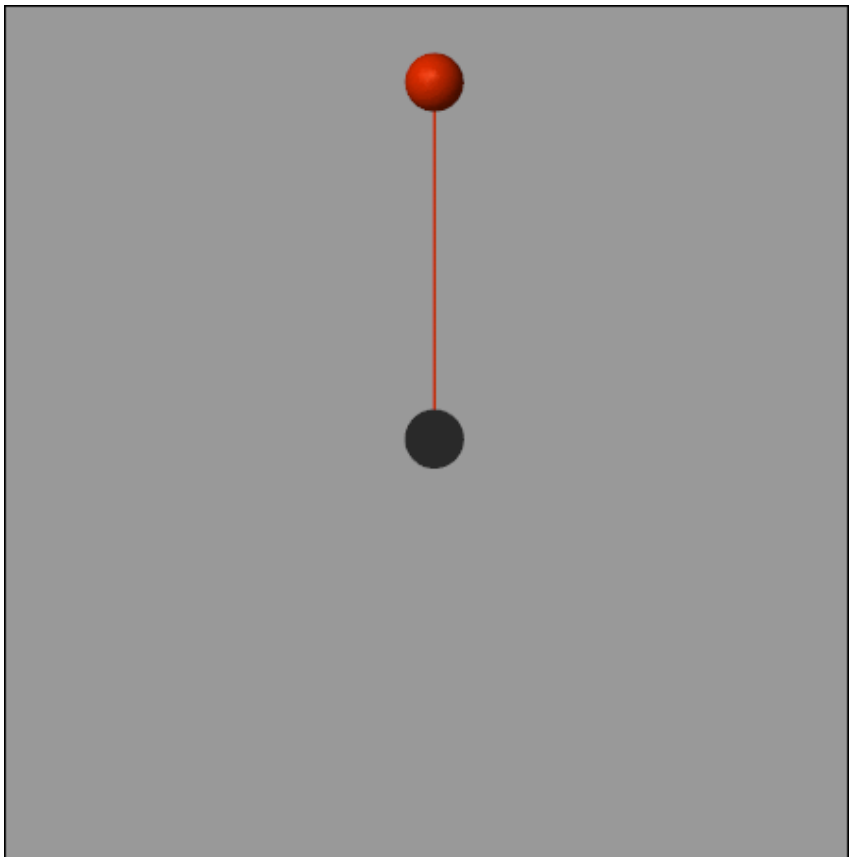




系统受干扰后能回到稳定状态，说明$(\theta, \omega) = (0, 0)$**稳定**

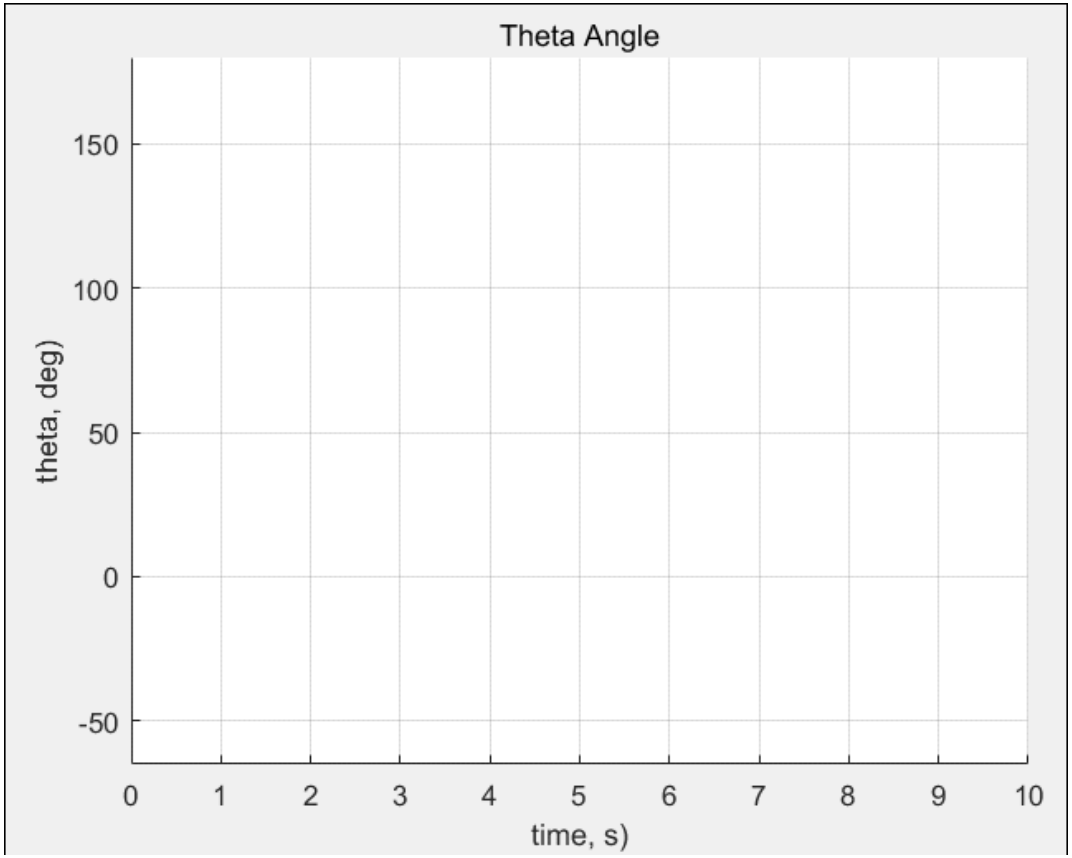
$(\theta, \omega) = (\pi, 0)$

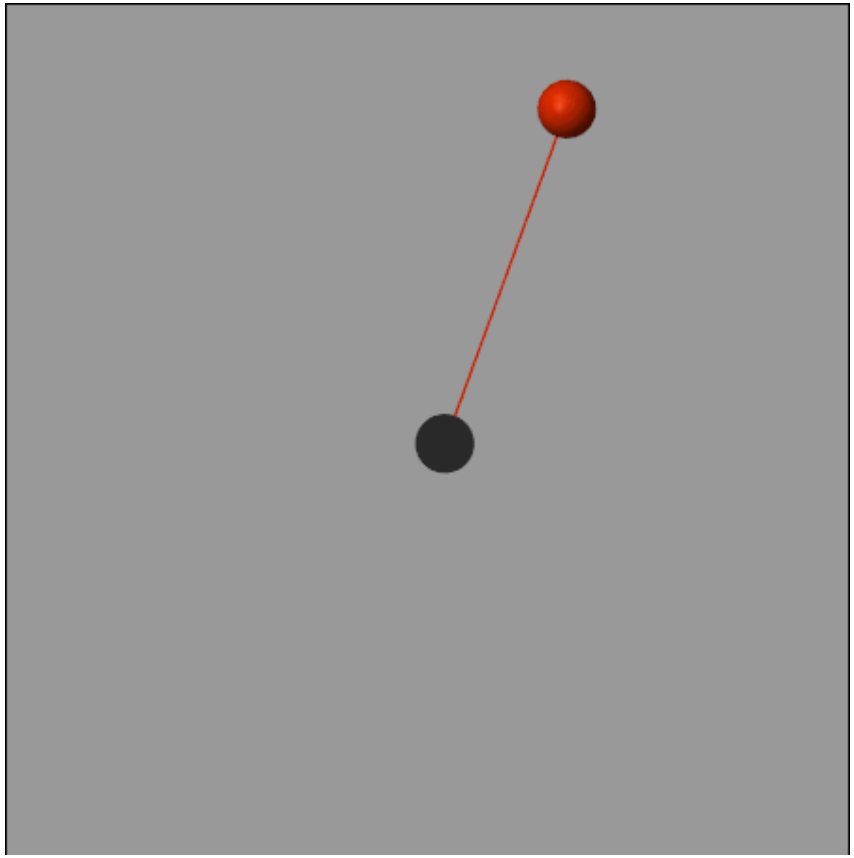




当系统不受干扰时，可以停在$(\theta, \omega) = (\pi, 0)$

$(\theta, \omega) = (\pi, 0)$





系统受干扰后偏离稳定状态，说明 $(\theta, \omega) = (\pi, 0)$**不稳定**

李雅普诺夫稳定性分析

李雅普诺夫稳定性直接判别法（充分条件）

$x = 0$是系统$\dot{x} = f(x)$的一个平衡点，李雅普诺夫函数$V: D \to \mathcal{R}$是连续可微分的函数。

- 若能找到一个V满足：

  1）V正定：$V(0) = 0$ 且 $V(x) > 0, \forall x \neq 0$

  2）$\dot{V}$半负定：$\dot{V}(x) \leq 0$，那么$x = 0$稳定。

- 进一步，若$\dot{V}$负定：$\dot{V}(x) < 0$，$\forall x \neq 0$，那么$x = 0$渐进稳定。
- 若能找到一个V满足$\dot{V}$正定，且以原点为中心的任意小球内V总能找到正值，那么$x = 0$不稳定。

钱学森, 宋健. 工程控制论[D]. , 1980. 第三版P105

李雅普诺夫稳定性分析

单摆稳定性分析

$$\begin{cases} \dot{\theta} = \omega \\ \dot{\omega} = -\dfrac{g}{L}\sin\theta + b\omega \end{cases} \xrightarrow[a = \frac{g}{L}]{\begin{matrix} x_1 = \theta \\ x_2 = \omega \end{matrix}} \begin{cases} \dot{x_1} = x_2 \\ \dot{x_2} = -a\sin x_1 + bx_2 \end{cases}$$

- 选择李雅普诺夫函数$V(x) = \frac{1}{2}\boldsymbol{x}^T P\boldsymbol{x} + a(1-\cos x_1)$，其中$P = \begin{bmatrix} 1 & b/2 \\ b/2 & b^2/2 \end{bmatrix}$
- 对$V(x)$求导可得：$\dot{V}(x) = -\frac{1}{2}abx_1\sin(x_1) - \frac{1}{2}bx_2^2$
- 当$0 < |x_1| < \pi$时，$x_1\sin(x_1) > 0$，即$\dot{V}(x)$负定
- 故$(x_1, x_2) = (0,0)$渐进稳定

---

http://users.isr.ist.utl.pt/~pedro/NCS2012/04_LyapStability.pdf

模型线性化

□ 局部线性化概念

当系统在**预定工作点**附近**小偏差**运行时，可以用线性模型来**近似**原非线性系统。目的是利用线性系统的特性，大幅简化计算。

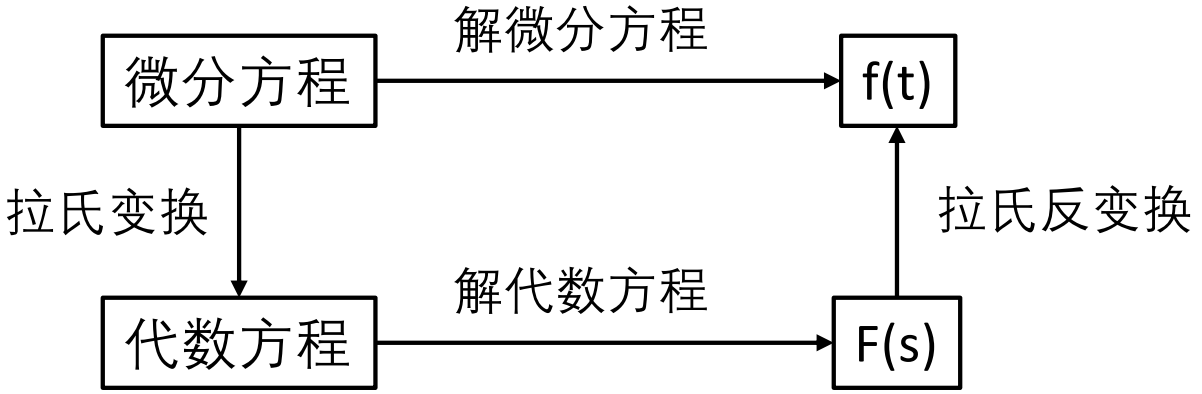
□ 单摆模型的线性化

在$(\theta,\omega)=(0,0)$点附近对非线性项进行一阶泰勒展开得$\sin\theta\approx\theta$，因此原非线性微分方程

$$\begin{cases}\dot{\theta}=\omega\\ \dot{\omega}=-a\sin\theta+b\omega\end{cases}\xrightarrow{\text{线性化为}}\begin{cases}\dot{\theta}=\omega\\ \dot{\omega}=-a\theta+b\omega\end{cases}$$

拉普拉斯变换

- **拉普拉斯变换法**(Laplace Transform，简称拉氏变换) 可以把**时域**的线性常系数**微分方程**转化为**频域**上的**代数方程**，大大降低求解难度。



- 拉氏变换、反变换通常可以查表计算。常用的变换公式：

$f(t)$	$F(s)$
单位脉冲 $\delta(t)$	1
单位阶跃 $1(t)$	$\frac{1}{s}$
指数函数 e^{-at}	$\frac{1}{s+a}$

拉普拉斯变换

- **拉氏变换微分定理**

$$L[f^n(t)] = s^n F(s) - s^{n-1} f(0) - \cdots - s f^{n-2}(0) - f^{n-1}(0)$$

- 系统在$(\theta, \omega) = (0, 0)$点可以线性化为 $\ddot{\theta} = -a\theta + b\dot{\theta}$
- 对其进行拉氏变换可得

$$s^2\Theta(s) - s\theta(0) - \dot{\theta}(0) = -a\Theta(s) + bs\Theta(s) - b\theta(0)$$

其中$L[\theta(t)] = \Theta(s)$，$\theta(0)$，$\dot{\theta}(0)$为初始时刻的状态

- 化简得到 $$\theta(s) = \frac{s\theta(0) + \dot{\theta}(0) - b\theta(0)}{s^2 - bs + a}$$
- 分母多项式$D(s) = s^2 - bs + a$称为系统的**特征多项式**

线性系统稳定性分析

1、时域分析法，求解微分方程

对系统在$(\theta,\omega)=(0,0)$处的微分方程 $\ddot{\theta}-b\dot{\theta}+a\theta=0$ 直接进行求解

特征方程　　$r^2-br+a=0$　　（r称为特征根）

齐次方程的通解　　$r_1=\frac{b+\sqrt{b^2-4a}}{2}$，$r_2=\frac{b-\sqrt{b^2-4a}}{2}$

由于$a>0,b<0$，故r_1和r_2的实部均小于0

解得θ的响应函数的形式有两种可能：

1）若r_1和r_2均为实数：　　$\theta(t)=C_1e^{r_1t}+C_2e^{r_2t}$

2）若$r_1=\alpha+\beta i, r_2=\alpha-\beta i$：　　$\theta(t)=e^{\alpha t}(C_1\cos\beta t+C_1\sin\beta t)$

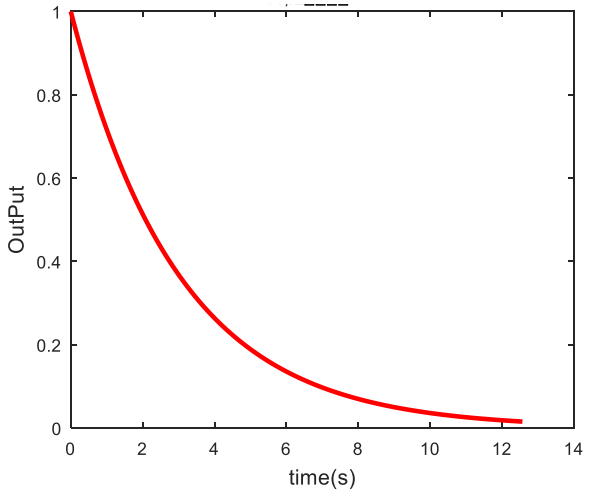
其中常数C_1和C_2由初始条件决定

线性系统稳定性分析

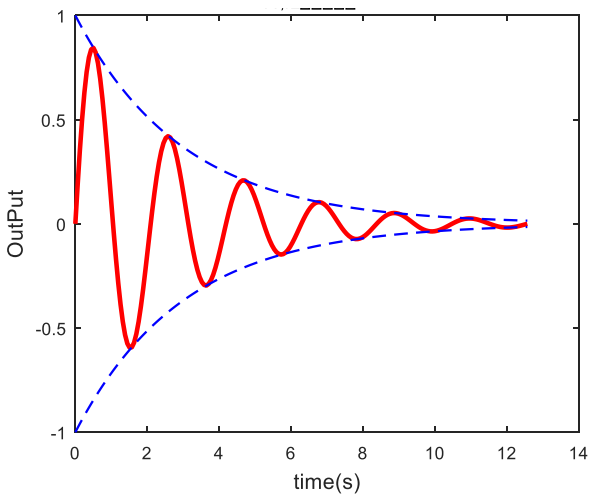
1、时域分析法，求解微分方程

r_1和r_2的实部均小于0时

1）$\theta(t) = C_1 e^{r_1 t} + C_2 e^{r_2 t}$



2）$\theta(t) = e^{\alpha t}(C_1 \cos\beta t + C_1 \sin\beta t)$



可以看出，响应渐进收敛，

系统在$(\theta, \omega) = (0, 0)$处是稳定的

线性系统稳定性分析

1、时域分析法，求解微分方程

对系统在$(\theta,\omega)=(\pi,0)$处的微分方程 $\ddot{\theta}-b\dot{\theta}-a\theta=-a\pi$ 进行求解

求暂态响应，只需要求齐次方程的通解

特征方程 $$r^2-br-a=0$$

通解为 $$r_1=\frac{b+\sqrt{b^2+4a}}{2},\quad r_2=\frac{b-\sqrt{b^2+4a}}{2}$$

由于$a>0,b<0$，故r_1的实部大于0

根据同样的分析，响应逐渐发散，系统在$(\theta,\omega)=(\pi,0)$处是不稳定的

线性系统稳定性分析

2、频域分析，特征多项式求根

极点：特征多项式的根

- 系统在$(\theta,\omega)=(0,0)$处的特征多项式$D(s)=s^2-bs+a$

  令$D(s)=0$，求得极点为 $s_1=\frac{b+\sqrt{b^2-4a}}{2}$，$s_2=\frac{b-\sqrt{b^2-4a}}{2}$

  由于$a>0,b<0$，极点的实部都小于0

  故极点在复平面上的位置均在虚轴的左半平面，此时系统稳定

- 同样，可以求出系统在$(\theta,\omega)=(\pi,0)$处特征多项式$D(s)=s^2-bs-a$的根，发现存在实部大于0的极点，此时系统不稳定

线性系统稳定性分析

- 除了以上两种方法判别稳定性，还可以用**劳斯判别法**
- 劳斯判别法不需要求解特征方程$D(s)$的根，只需根据$D(s)$的系数，通过**代数计算**直接判别根的位置
- 由劳斯判据可以推导出**二阶系统**的稳定性判定条件：

  对$D(s) = c_2 s^2 + c_1 s + c_0$，若所有系数满足$c_n > 0$，则线性系统稳定
- 对单摆的特征多项式 $D(s) = s^2 + a - bs$，由于$a > 0, b < 0$，能满足以上条件，故单摆在$(\theta, \omega) = (0, 0)$处稳定

https://zh.wikipedia.org/zh-hans/劳斯–赫尔维茨稳定性判据

线性系统稳定性分析

以下两种稳定性判别方式等价：

- 极点的位置（频域）

稳定：所有极点的实部都小于0，即全部极点位于复平面左半平面

不稳定：只要有一个极点位于复平面右半平面

临界稳定：有部分极点位于虚轴上，其余位于左半平面

- 求解微分方程（时域）

稳定：所有通解的实部都小于0

不稳定：只要有一个通解的实部大于0

案例2 - 倒立摆

由于倒立摆本身不稳定，加入**控制输入**u来保持倒立摆垂直

物理模型

↓

（非线性）微分方程模型

↓

线性化

↓

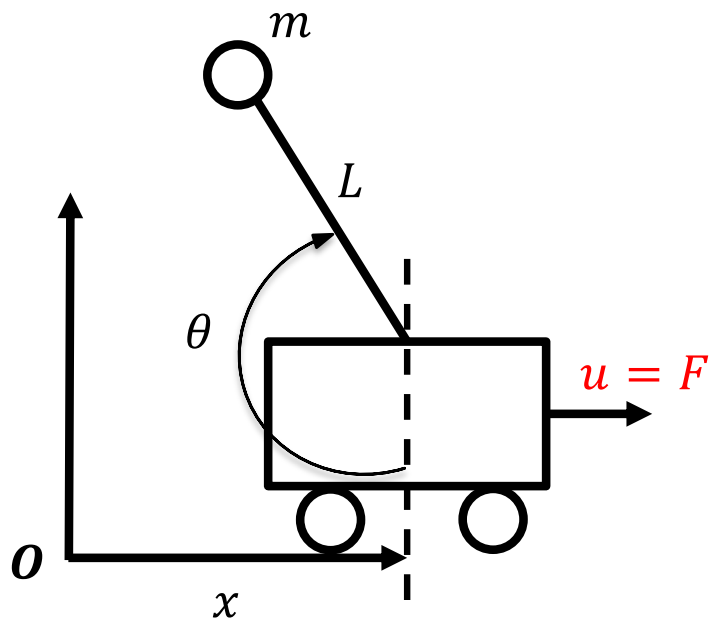
线性系统稳定性判定

↓

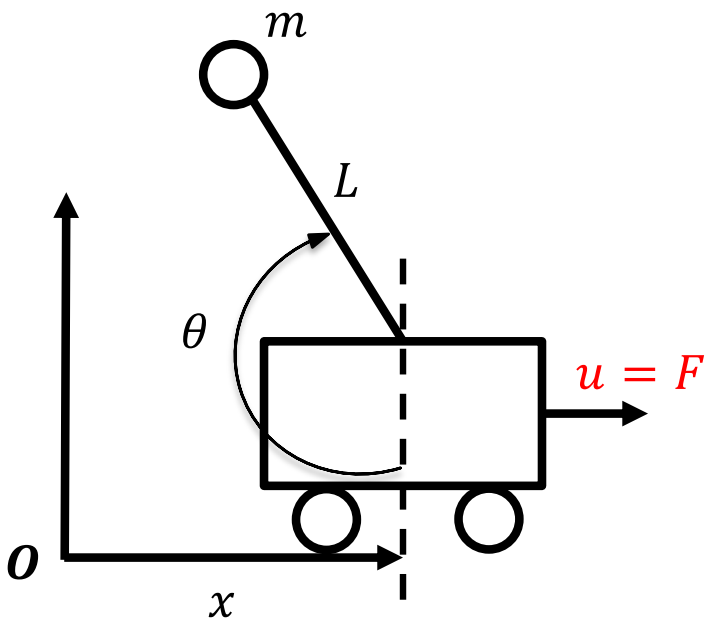
拉氏变换、传递函数

↓

控制器设计



动力学建模



g：重力加速度

d：地面摩擦系数

L：摆杆长度

u：控制输入

x：小车水平距离

m：摆球质量

M：小车质量

$$(M + m\sin^2\theta)\ddot{x} = -mg\cos\theta\sin\theta + mL\dot{\theta}^2\sin\theta - d\dot{x} + u$$

$$L\ddot{\theta} = g\sin\theta - \ddot{x}\cos\theta$$

https://en.wikipedia.org/wiki/Inverted_pendulum Inverted pendulum on a cart

状态方程

- 倒立摆的**系统状态方程**

对微分方程在$\theta = \pi$点附近线性化，有

$$\begin{cases} M\ddot{x} = -mg(\pi - \theta) - d\dot{x} + u \\ L\ddot{\theta} = -g(\pi - \theta) + \ddot{x} \end{cases}$$

选择状态变量$\overline{\boldsymbol{x}} = \left[x \;\; \dot{x} \;\; \theta \;\; \dot{\theta}\right]^T$，系统状态方程可写为

$$\dot{\overline{\boldsymbol{x}}} = A\overline{\boldsymbol{x}} + Bu$$

其中$A = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & -\frac{b}{M} & -\frac{mg}{M} & 0 \\ 0 & 0 & 0 & 1 \\ 0 & -\frac{b}{ML} & -\frac{(m+M)g}{ML} & 0 \end{bmatrix}, B = \begin{bmatrix} 0 & \frac{1}{M} & 0 & \frac{1}{ML} \end{bmatrix}^T$

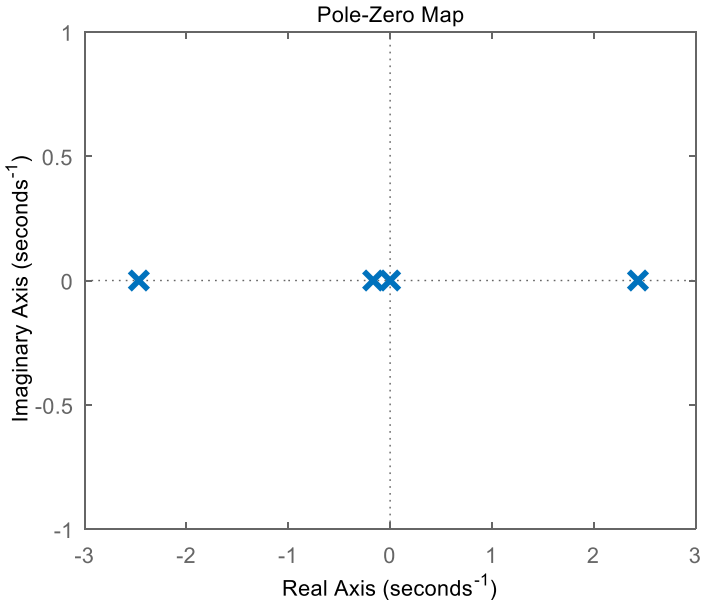
控制器设计

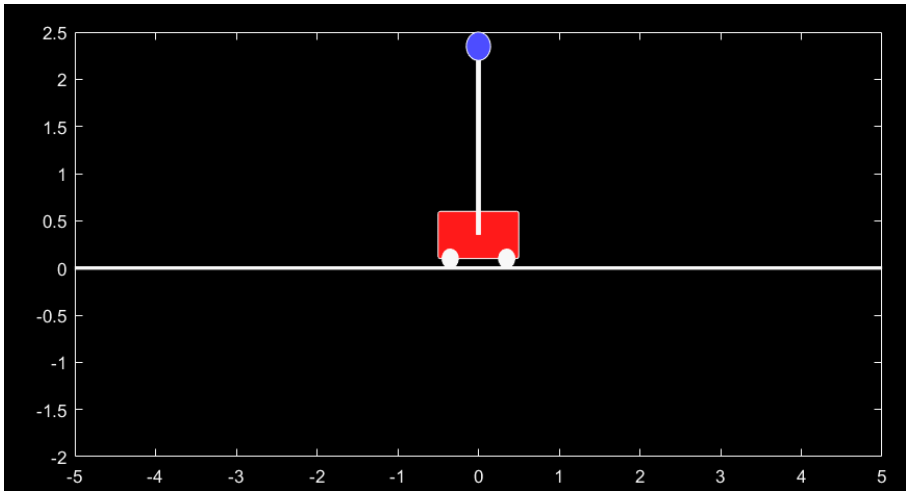
- 开环极点位置

求极点，matlab命令：

```
SYSTEM = sys(A,B,C,D)
pzplot(SYSTEM)
```

matlab仿真程序：cartpend_code/sim_cartpend.m





右半平面含有极点，故开环系统不稳定

极点配置

- 线性常系数系统的**稳定性**取决于极点是否位于左半平面
- 系统其他**特性和品质**也与极点在左半平面上的位置有关

——可以根据系统设计指标要求，有目的地配置极点分布

- 极点是否可以通过反馈任意配置？

  在什么情况下可以这样配置？

钱学森, 宋健. 工程控制论[D]. , 1980. 第三版P168

极点配置

- 当系统**完全可控**时，极点可以用状态线性反馈任意配置
- **可控性**判定定理

设状态量$\overline{\boldsymbol{x}} \in \mathcal{R}^n$，那么对线性系统

$$\dot{\overline{\boldsymbol{x}}} = A\overline{\boldsymbol{x}} + Bu$$

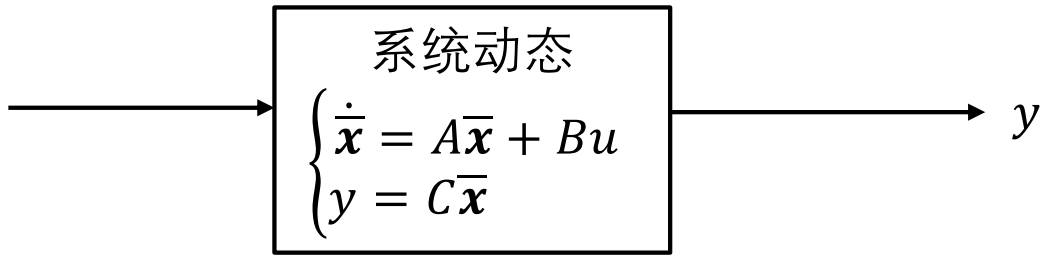
若**可控性矩阵** $M = [B \;\; AB \;\; A^2B \;\; \ldots]$的秩为$n$，则系统可控

- matlab命令：

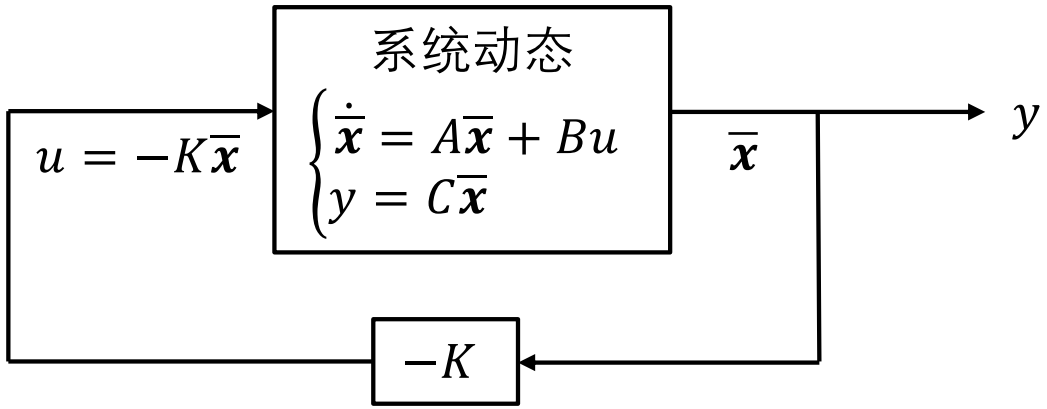
```
M = ctrb(A,B)
rank(M)
```

极点配置

- **开环系统**



- **闭环系统** 假设$\bar{x}$均可测量，进行**状态反馈**



闭环系统变为：$\dot{\bar{x}} = (A - BK)\bar{x}$

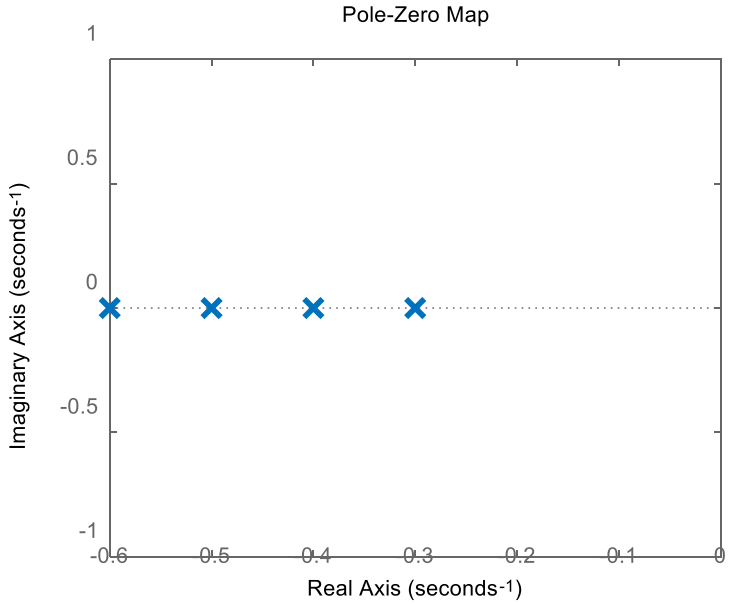
因此可以通过设计K来将系统极点配置到任意位置

极点配置

第一种情况：缓慢

仿真程序：cartpend_code/poleplace_cartpend.m

```
p = [-.3; -.4; -.5; -.6];
K = place(A,B,p);
```



极点都在左半平面，闭环系统稳定

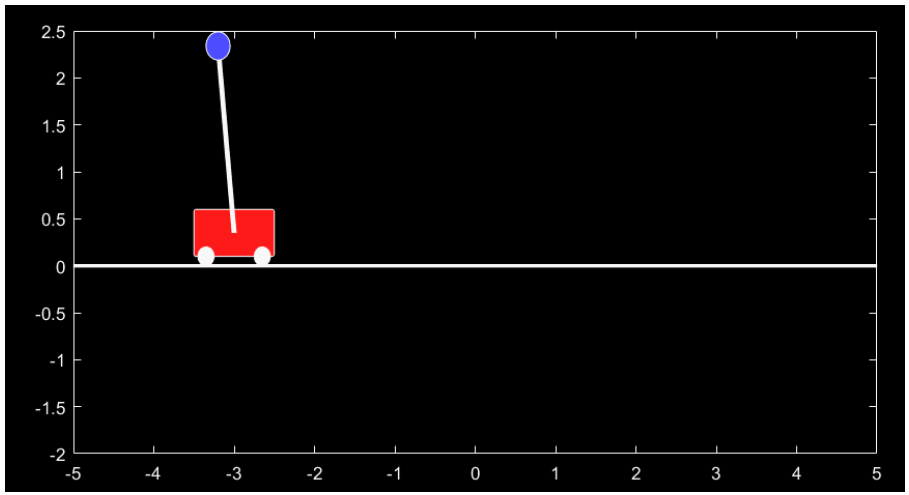
负实部绝对值较小，收敛较慢

解得$K = [-0.04, -1.34, 71.97, 18.68]$

利用线性反馈 $u = -K\bar{x}$ 控制倒立摆

初始状态 $\bar{x}_0 = [-3 \quad 0 \quad \pi + 0.1 \quad 0]^T$

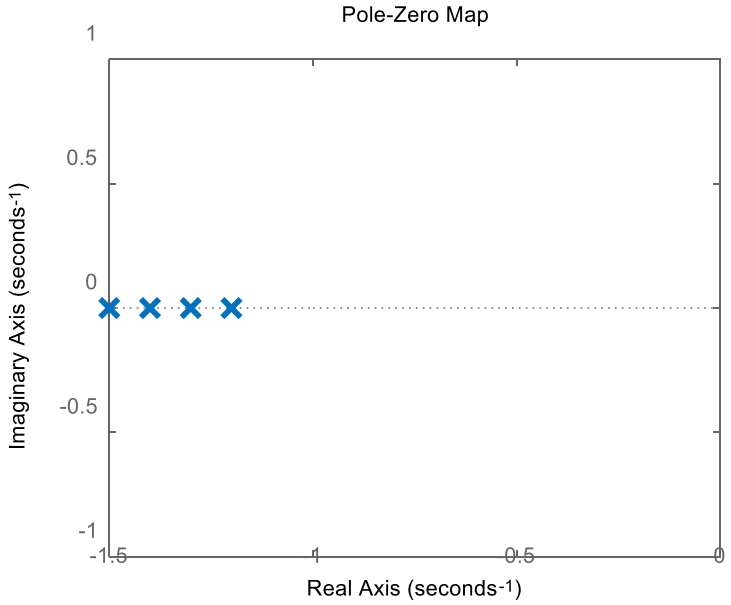
目标状态 $\bar{x}_{cmd} = [1 \quad 0 \quad \pi \quad 0]^T$



极点配置

- 第二种情况：适中

```
p = [-1.2;-1.3;-1.4;-1.5];
K = place(A,B,p);
```



极点都在左半平面，闭环系统稳定

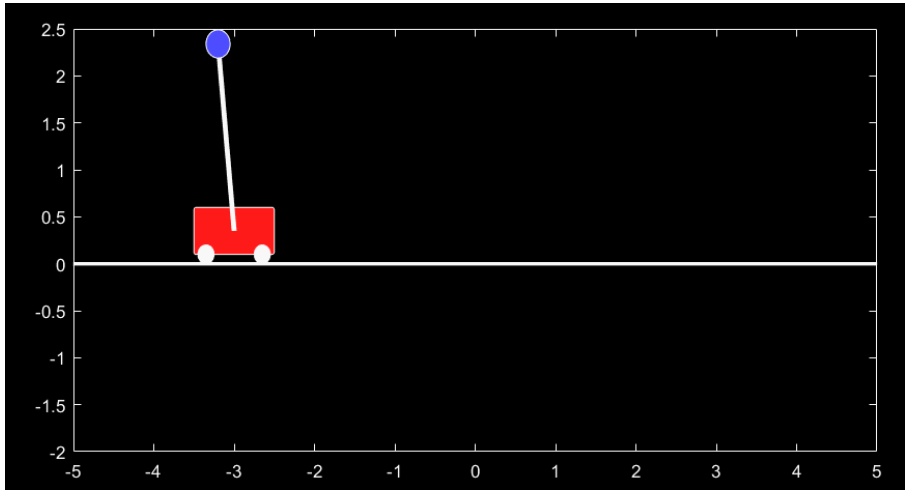
负实部绝对值较大，收敛较快

解得$K = [-3.3, -10.8, 175.7, 73.5]$

利用线性反馈 $u = -K\bar{x}$ 控制倒立摆

初始状态 $\bar{x}_0 = [-3 \quad 0 \quad \pi + 0.1 \quad 0]^T$

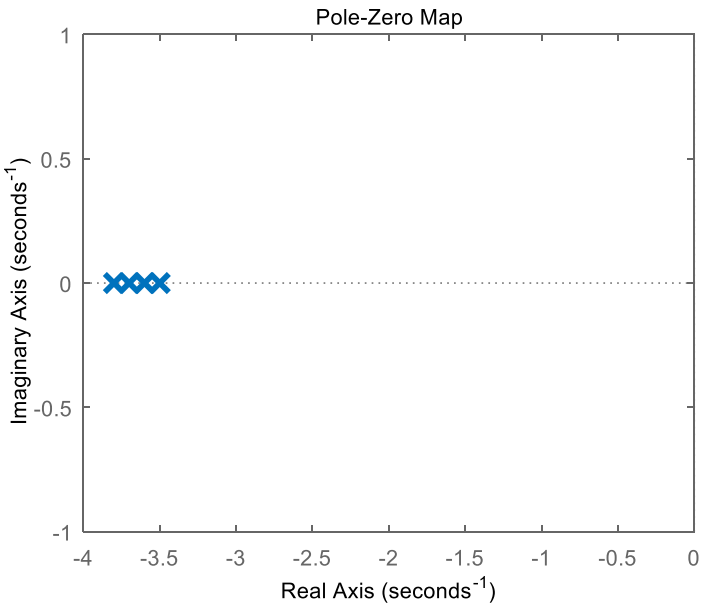
目标状态 $\bar{x}_{cmd} = [1 \quad 0 \quad \pi \quad 0]^T$



极点配置

- 第三种情况：不稳定

```
p = [-3.5;-3.6;-3.7;-3.8];
K = place(A,B,p);
```



此时若继续向左移动极点，将

解得$K = [-177, -195, 1213, 534]$

控制 $u = -K\overline{x}$ 会出现剧烈抖动，

甚至超出系统极限；

且原非线性系统将变得不稳定。

➔**并非真的可以“任意”配置极点**！